Excitation-Aware On-Track System Identification at Full Scale: A Simulation Study

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***Abstract*—On-track system identification promises to learn tire models from ordinary racing laps instead of dedicated steady-state manoeuvres. We study one such residual-learning pipeline at full vehicle scale, in simulation, on a plant that publishes its own per-wheel slip angles, loads and forces as ground truth. Transplanted unchanged from a small-scale platform, it converges on a tire with less than half the vehicle's grip ceiling and most Pacejka coefficients on their box constraints—not a solver, bounds or tuning failure but a loss of excitation created by the pipeline's own virtual-data step. The friction the synthetic rollout can demand, $\mu_{\mathrm{reach}} = v^2\delta_{\mathrm{sweep}}/(gL)$, is a property of the rollout configuration, not of the tire. We verify that on a grid over rollout speed, sweep amplitude and true peak factor with a known tire: below the tire's own peak the fit returns the excitation rather than the tire, whatever the solver and start point. We then correct the pipeline—the rollout speed taken from the measured lap, the sweep bounded to the observed steering range, the regularisation target frozen—and gate the model-to-controller boundary on derived physical quantities rather than on per-coefficient bounds. Over three timed laps no identified coefficient rests on a bound and the corrected pipeline preserves a peak factor within 0.05 of the plant's effective friction, against 0.77 for the inherited configuration—though that factor is carried by the prior, not recovered from the lap. All results are in simulation, one run per configuration.**

***Index Terms*—Autonomous vehicles, system identification, parameter estimation, vehicle dynamics, tires, predictive control, road vehicles, simulation.**

I. Introduction

MODEL-BASED controllers for autonomous racing derive their performance from the fidelity of the vehicle model they carry [1], [2], [3]. In the lateral channel that fidelity is dominated by the tire, which is simultaneously the physical bound on manoeuvrability and the least accessible component to measure: its behaviour depends on surface, temperature, wear and load in ways that change between sessions [4], [5].

Classical characterisation resolves this by removing the vehicle from the track—steady-state circular or ramp-steer manoeuvres in a large open space yield clean data at a logistical cost a race weekend rarely affords [6]. On-track identification removes the logistics but reintroduces transients, noise and, critically, the fact that a racing line is a poor excitation signal for a nonlinear model; nonlinear least squares applied directly to lap data is known to be brittle in exactly this regime [6].

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Dikici *et al.* [6] propose the hybrid remedy that is the starting point of this paper. A small residual network is trained on 30 s of ordinary lap data to predict the one-step error of a nominal single-track model; the corrected model is then rolled out through a *synthetic*, quasi-steady steering sweep; and Pacejka coefficients are fitted to that synthetic sweep. Because the sweep is generated rather than driven it can be made steady and noise-free, and because the residual network only has to interpolate within its training distribution its generalisation weakness is contained. The procedure iterates, each fit becoming the next nominal model. On a 1:10 platform it reports a 3.3× lower one-step RMSE than nonlinear least squares under noise, from 30 s of driving [6].

A. Problem statement

This paper asks what the pipeline identifies when it is moved from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase, 16 ° of road-wheel lock, following a reference speed profile of 9.6–25.8 m/s and achieving 9.2–26.5 m/s in the identification data. The question is one of identifiability rather than of parameter transfer. Two scale-dependent mechanisms, both latent at 1:10, break the estimate:

1) **Excitation collapses inside the virtual-data step.** The Pacejka fit never sees the recorded lap; it sees the synthetic rollout, whose reachable friction $\mu_{\mathrm{reach}} \approx v^2\delta_{\mathrm{sweep}}/(gL)$ is a property of the rollout configuration, not of the tire. A rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire's peak the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$: only the product BCD is identifiable and the bounded optimiser resolves the remaining freedom on a box edge [7], [4].
2) **Lateral demand scales as v^2, so a bad grip estimate is silent until it is catastrophic.** Driven through the same controller on an earlier, faster reference trajectory, one identified set with a peak friction coefficient of 0.40 tracked at 0.16 m of cross-track error at 11.1 m/s, 0.43 m at 15 m/s and 41.89 m at 27 m/s. The scaled platform never reaches the speed at which the error becomes observable. The figures are from an offline closed-loop harness rather than from a controlled simulator run and are reported as such throughout (Section VII-A).

A third difference is an opportunity. Because the study is conducted in the CARLA simulator [8] through a purpose-built

ROS 2 interface, the simulator's own per-wheel telemetry—slip angle, normal load, lateral force, per wheel—is available as ground truth, so an on-track identifier can be scored against the forces the plant actually generated.

B. Contributions

The contributions are an identifiability result, the corrections it implies, the gate that keeps an unidentified model out of the controller, and the architecture that makes all three measurable against the plant's own forces.

1) **An identifiability analysis of the virtual-data step** (Section V-B), giving the closed-form reachable-friction bound above and an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns whenever the configuration cannot reach the prior's peak. The bound is verified directly in Section VI-B, on a grid over rollout speed, sweep amplitude and true peak factor with a known tire: below the tire's own peak the fit returns the excitation rather than the tire, at every grip level tested and independently of the solver. This corrects the baseline method [6] at any scale; it is latent at 1:10 because a 2–4 m/s rollout is representative there and is not here.
2) **The corrections the bound implies**, on both sides of the fit. On the rollout side (Section V-C): the synthetic sweep is bounded at $1.5\times$ the 99th percentile of the observed steering, so the fit cannot read the residual network's extrapolation as grip, and the Tikhonov target is frozen at the configured prior while the optimiser's start point tracks the iterations—together removing a drift that took D from the 1.009/1.002 front/rear prior to 1.5402 and 1.8721 over six iterations. On the measurement side (Section V-E): slip angles built from the *achieved* road-wheel angle rather than the command, mass and wheelbase from measured static wheel loads rather than from a configuration file, and a friction warm-start read from the *curvature* of the axle force curve and applied as a floor rather than as a median over a low-slip mask.
3) **Admissibility gates on derived physical quantities** (Section VII) at the model-to-controller boundary—axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the trajectory's lateral demand—rather than per-coefficient box checks, which cannot detect a degenerate fit or a pairwise-oversteering axle set.
4) **A simulation architecture that supplies the ground truth, and a closed-loop evaluation on it** (Sections IV-B–IV-E and VI). CARLA_ASU_BRIDGE exposes the plant's own per-wheel forces, slip angles and loads as ROS 2 topics alongside a conventional autonomy interface, so an on-track identifier can be scored against the forces the plant generated. On a static 1.05 friction surface over three timed laps the corrected pipeline *preserves* a peak factor within 0.05 of the plant's effective friction where the inherited configuration starts and ends on its box constraint, and halves the lateral tracking error under MPC. On this trajectory that factor is carried by the configured prior rather than recovered from the lap—the accepted fits return the prior to four decimal places—and Section VI-A4 states where the prior came from. We report the two axes on which the inherited configuration is better, and the reason: the lap reaches 2.2 ° of slip, where a wrong peak factor costs nothing.

The paper follows that order: Section II places the work against the state of the art; Section III states the identification problem and its practical-identifiability properties; Section IV describes the simulation architecture and data pipeline; Section V the identification methodology; Section VI the results; Section VII how the identified model is handed to the controller; and Sections VIII–IX the limitations and outlook. All results are obtained in simulation, and Section VIII states which conclusions are expected to transfer to hardware.

II. RELATED WORK AND POSITION

A. Tire models and Magic Formula identification

The Magic Formula of Bakker, Nyborg and Pacejka [9] and its consolidation [10], [4] remains the standard empirical description of steady-state tire force generation and is the model identified here; its lateral form and the two properties this paper relies on are stated in Section III. Classical characterisation obtains the coefficients from steady-state manoeuvres in a controlled space [6], [5]: clean data at high logistical cost. Direct fitting to measured contact-patch forces is the most informative variant and requires a force-measuring hub or a simulator's own wheel telemetry; in this work it is used only as a *reference* against which the on-track identifier is scored, never inside the identification loop. Between the two sits interactive identification from track or rig data, of which TRIP-ID [11] is representative: an operator-guided fit of the full Magic Formula to recorded manoeuvres. The pipeline studied here targets the same output with no operator and no dedicated manoeuvre, which is precisely what puts its excitation—and therefore its identifiability—outside the engineer's control.

B. On-track and data-driven identification

On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical baseline and is brittle under noise [6]. The recursive alternative to a batch fit is a dual or joint filter that carries states and parameters together—Wenzel *et al.* [12] is the canonical vehicle instance, with Sierra *et al.* [13] the lateral-dynamics cornering-stiffness variant. The recursive form is subject to the same excitation condition this paper is about. A filter run on a smooth racing line has as little information about the tire's peak as a batch fit of the same data; it merely expresses that as a parameter covariance that does not shrink, rather than as a coefficient on a bound. Dedicated friction estimators make the excitation dependence explicit [14], [15], [16], [17], and Hsu *et al.* [18] is the closest prior art to the warm start of Section V-E3: it recovers the friction limit below the peak from the *curvature* of the force response, read through the pneumatic trail in steering torque. Our estimator uses the same observation on a different signal: the axle force curve reconstructed from the IMU and odometry, with no steering-torque sensor and no assumption about the steering system. It is then gated and applied as a one-way floor—a property of the interface to the controller rather than of the estimator. Gaussian-process approaches learn the mismatch of a nominal model and have been used both for identification and inside the controller [19], [20], [2]; they are accurate but carry a continuing computational cost and, as [6] observes, are usually evaluated starting from an already accurate physical model. Neural residuals replace the Gaussian process with a cheaper regressor at the cost of generalisation guarantees.

Deep Dynamics [21] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer clamping the internal estimates into nominal ranges. Our admissibility gates share the motivation but differ in placement and in what is constrained. We report as a substantive finding that per-coefficient box constraints are *insufficient*: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise over-steering and open-loop unstable above 14.5 m/s. We therefore gate on derived quantities at the model-to-controller interface and treat a coefficient resting on a bound as a diagnostic of non-identifiability rather than as a successful clamp.

The hybrid residual-plus-virtual-data structure of [6] is the pipeline we extend. Its key idea—query the learned model only where it interpolates, and extract physically parameterised coefficients from it—is retained unchanged. What we correct is the mechanism that decides *where* the learned model is queried.

C. Overlap with concurrent work

Wu *et al.* [22] extend the same baseline concurrently and independently, introducing a vision-based friction prior, an S4 structured state-space residual in place of the MLP, and Nelder–Mead for the iterative Pacejka extraction. The pipeline described here independently contains an S4D temporal residual [23], [24], a friction warm-start, and Nelder–Mead and differential-evolution solver options, so we claim none of the three as contributions. Two differences are technical rather than chronological: our friction prior is proprioceptive, so it needs no camera or textured-surface assumption, and it is read from the *curvature* of the axle force curve by a gated brush-model fit [25], [16] rather than from utilised friction, which on a gentle lap measures 0.044 median on this vehicle and is a lower bound on what is available (Section V-E3d). Neither prior is the fix: the degeneracy is located in the rollout excitation, not in the initialisation, and to our knowledge the reachable-friction bound of Section V-B and its consequence for the virtual-data step have not previously been stated.

D. Simulation architectures for identification and validation

CARLA [8] is the standard open simulator for autonomous-driving research and supplies the sensor and scenario infrastructure used here. Its native dynamics come from the underlying PhysX wheeled-vehicle model [26]—a saturating friction-and-stiffness tire law (Section IV-C) rather than an empirical Magic Formula fit to measured data—which is why this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth Pacejka coefficients. Recent work addresses that fidelity gap directly: R-CARLA [27] makes the dynamics model interchangeable while retaining CARLA's sensor simulation, and Sagmeister *et al.* [28] quantify how far a vehicle model can be simplified before conclusions about a trajectory-following controller stop transferring—finding, relevantly here, that sensitivity is largest near the acceleration limits. CARLA has also hosted complete Formula Student Driverless stacks with ROS 2 interfaces [29]. The architecture of Section IV sits in this line and is distinguished by exposing the simulator's own per-wheel slip, load and force telemetry on a simulation-time-stamped interface, so that an identification method can be scored against the forces the plant actually produced rather than against a secondary model. Scaled platforms—F1TENTH [30] and the ForzaETH Race Stack [3]—provide the software context of the baseline; full-scale Formula Student Driverless systems [5] the vehicle class targeted here.

E. Downstream consumer and gap addressed

The identified model is consumed by a nonlinear MPC path-tracking controller built on `acados` [31] with partial-condensing HPIPM [32], following standard practice in racing MPC [1], [2], tracking a minimum-curvature raceline generated in the manner of Heilmeier *et al.* [33]. We claim no contribution to the controller itself; the controller appears in this paper only as the consumer that defines what an admissible identification is (Section VII).

Summarising the gap: the baseline [6] establishes that a residual network plus virtual steady-state data recovers a Pacejka model from a lap at 1:10 scale, and concurrent work [22] improves its residual architecture and its initialisation. Neither addresses the condition under which the virtual data itself carries information about the tire's peak, which is the binding constraint at full scale, and neither treats the admissibility of the identified model for its downstream controller as an explicit, physically derived contract. This paper addresses both, on a platform where the answer can be checked against the plant's own forces.

III. Identification Problem

A. Model structure

The identification model is the dynamic single-track model of the baseline, with mass m, yaw inertia I_z and centre-of-gravity distances l_f, l_r, $L = l_f + l_r$. Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \tfrac{1}{m}\big(F_{y,r} + F_{y,f}\cos\delta - m v_x \omega\big), \tag{1}$$

$$\dot{\omega} = \tfrac{1}{I_z}\big(F_{y,f} l_f \cos\delta - F_{y,r} l_r\big), \tag{2}$$

with slip angles

$$\alpha_f = \delta - \arctan\Big(\tfrac{v_y + l_f\omega}{v_x}\Big), \quad \alpha_r = -\arctan\Big(\tfrac{v_y - l_r\omega}{v_x}\Big). \tag{3}$$

Axle forces follow the lateral Magic Formula

$$F_y = F_z\, D \sin\!\big(C \arctan\!\big(B\alpha - E(B\alpha - \arctan B\alpha)\big)\big), \quad (4)$$

with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The identification state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$, and the nominal one-step map is $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. The baseline discretises (1)–(2) by explicit Euler at T_s; we retain Euler as the default and expose second- and fourth-order Runge–Kutta as options, because the full-scale vehicle's lateral mode is stiffer relative to T_s than the scaled platform's.

B. Objective and derived quantities

The identification objective is to recover Φ_p from a single 30 s segment of ordinary lap data, without dedicated manoeuvres, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities rather than on coefficients. The linearised axle cornering stiffness is the slope of (4) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\mathrm{MF}} D_i, \qquad i \in \{f, r\}, \quad (5)$$

(writing C^{MF} for the shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\max} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \quad (6)$$

The linear model built from (5) is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\mathrm{crit}} = L\sqrt{C_f C_r / \big(m(l_f C_f - l_r C_r)\big)}$ [34].

C. Practical identifiability under limited excitation

Two properties of (4) are used throughout. First, its slope at zero slip is $F_z BCD$: the cornering stiffness depends on the coefficients only through that product, and E drops out. Second, and decisively, if the data does not approach the peak then (4) linearises to $F_y \approx F_z(BCD)\alpha$ and B, C, D become mutually unidentifiable. The distinction matters for what is being claimed. Given data that reaches the peak, (4) is *structurally* identifiable in the sense of Bellman and Åström [35]: the map from coefficients to noise-free outputs is injective, and no amount of excitation design is needed to make it so. What fails here is *practical* (a-posteriori) identifiability [36]: under a specific excitation the likelihood is flat along the BCD direction, so the parameters are not recoverable from the data at hand even though the structure admits them. That is the standard persistent-excitation condition [7], [37] expressed on this model, and neither the structural result nor the excitation condition is claimed as novel. What we claim, and demonstrate in Section V-B, is that the baseline pipeline's virtual-data step can silently violate it, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge—so the violation is reported as a converged fit rather than as a failure.

D. Baseline pipeline

The baseline proceeds in four stages, repeated for N_{it} iterations [6]:

1) **Residual learning.** A small network f_{NN} is trained on the recorded lap to predict the one-step error $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ from $[v_x, v_y, \omega, \delta]_k$; the corrected model is $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1} + e_k$.
2) **Virtual steady-state generation.** The corrected model is integrated open-loop at constant longitudinal speed while the steering angle is ramped from 0 to δ_{sweep}, producing a synthetic quasi-steady dataset.
3) **Pacejka fit.** Assuming $\dot{v}_y = \dot{\omega} = 0$ on the synthetic data, axle forces are recovered from the yaw balance,
$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \qquad F_{y,f} = \frac{m l_r v_x \omega}{L \cos\delta}, \quad (7)$$
slip angles from (3), and Φ_p is fitted to $(\alpha_i, F_{y,i})$ by bounded least squares.
4) **Re-nomination.** The fitted Φ_p becomes the next iteration's nominal model and the residual network is reinitialised.

The point that drives the rest of this paper is that stage 3 *never sees the recorded lap*. It sees the output of stage 2. The recorded lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

IV. Simulation Architecture and Data Pipeline

The identification pipeline is exercised end to end inside a single simulation architecture: a Formula Student vehicle model in CARLA acting as the plant, a C++ ROS 2 communication layer that drives it and republishes its state, and an identification stack that consumes that state. The architecture is described here because two of its properties—per-wheel ground truth and simulation-time stamping—are what make the identifiability claims of Section V checkable rather than merely arguable.

A. Choice of simulator

CARLA [8] was selected against four requirements that the identification problem of Section III imposes on a simulator.

Per-wheel ground truth. The claim to be checked is that the recovered Magic Formula reproduces the lateral force the plant actually produced, so the simulator must expose per-wheel slip angle, normal load and tire force—not only chassis states. CARLA's PhysX wheeled-vehicle solver computes these quantities internally and its client API exports them, which is what makes the tire forces feedback channel of Section IV-D possible.

Deterministic, simulation-time-stamped execution. Persistent-excitation and residual-learning arguments are only checkable if every sample carries a consistent clock. CARLA's synchronous mode with a fixed physics step (0.033 s here) decouples data timing from wall-clock load; all identification data is stamped from the simulation clock (Appendix, Section A7).

TABLE I
VEHICLE CONFIGURATION AND MEASURED PLANT PROPERTIES.

Quantity	Value	Unit
Body mass (configured)	240.0	kg
Static wheel load (total)	2644.5	N
Total (kerb) mass m[a]	269.6	kg
Unsprung mass[b]	29.6	kg
Front/rear static split	51.2 / 48.8	%
Wheelbase L	1.5334	m
l_f / l_r	0.7381 / 0.7953	m
Yaw inertia I_z[c]	51.1	$\mathrm{kg\,m^2}$
Wheel radius	0.25	m
Max road-wheel angle	16.0	°
Configured wheel friction	1.5	—
Road-surface factor	0.70	—
Effective friction[d]	1.05	—
Realised peak axle friction	1.00–1.05	—
Drag coefficient	0.426	—

Configured values are set in the bridge's `carla_interface_config.yaml`; measured values are obtained from the simulator's per-wheel telemetry. [a]Total static wheel load divided by $g = 9.81\,\mathrm{m/s^2}$ ($2644.5/9.81 = 269.6$). This is the mass every equation in this paper uses, not the configured body mass. [b]Total minus body mass; not measured independently. [c]Assumed; see Section A. [d]Configured wheel friction multiplied by the road surface's own coefficient, as reported per wheel by `feedback/tire_forces`. It is the effective value, not the configured one, that the identifier is scored against (Section V-E3d).

Full-scale vehicle and racing-track support. The target class is a full-scale Formula Student car on a closed circuit. CARLA accepts custom vehicle assets—here the Formula AI vehicle on the `silverstone` map—and offers a client that the bridge links against directly. Surveys of open self-driving simulators place CARLA at the state of the art for end-to-end stack testing on exactly these grounds [38].

B. Plant: Formula AI vehicle in CARLA

The plant is the driverless Formula Student–class vehicle model, instantiated in CARLA [8] (server 0.9.16, Unreal Engine 4.26) on the `silverstone` map: a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle whose lateral forces are generated by the PhysX wheeled-vehicle tire model [26] under a configured friction coefficient and lateral stiffness, not by a Magic Formula. The simulator therefore has no "true" Pacejka coefficients to recover; instead, the lateral forces PhysX itself computes at each wheel are taken as ground truth, and the identified Magic Formula is validated by comparing its predictions against those forces rather than against any coefficient the simulator holds (Section V-E).

C. The plant's tire curve, in closed form

The identification target is worth stating exactly, because it is knowable here. PhysX computes the lateral force of each wheel with `PxVehicleComputeTireForceDefault` [26]; at zero longitudinal slip and zero camber that expression collapses to

$$F_y(\alpha) = \mathrm{sgn}(\alpha)\,\mu F_z\,S_1(K), \qquad K = \frac{C_{\mathrm{lat}}|\tan\alpha|}{\mu F_z}, \tag{8}$$

with $C_{\mathrm{lat}} = F_z^{\mathrm{rest}}\,k_y\,S_1\big(3\bar{F}_z/k_x\big)$, $\bar{F}_z = F_z/F_z^{\mathrm{rest}}$ the normalised load, (k_y, k_x) the wheel's lateral-stiffness parameters, and $S_1(K) = \min(1,\, K - K^2/3 + K^3/27)$ the engine's smoothing function.

TABLE II
THE PLANT'S LATERAL TIRE CURVE: THE PARAMETERS OF (8), READ FROM THE SIMULATOR'S OWN WHEEL CONFIGURATION AND TELEMETRY.

Quantity	Value	Unit
Effective friction μ	1.05	—
Lateral stiffness k_y	17.0	/rad
Max normalised load k_x	2.0	—
Initial slope C_{lat} at $\bar{F}_z = 1$	$14.875\,F_z^{\mathrm{rest}}$	N/rad
front axle, $F_{z,f} = 1373\,\mathrm{N}$	20.4	kN/rad
rear axle, $F_{z,r} = 1274\,\mathrm{N}$	19.0	kN/rad
Peak reached at, $\bar{F}_z = 1$, $\mu = 1.05$	12.0	°

The initial slope is set by the *rest* load and by the normalised load through S_1, not by F_z alone: $C_{\mathrm{lat}} = k_y S_1(3\bar{F}_z/k_x)\,F_z^{\mathrm{rest}}$ evaluates to $14.875\,F_z^{\mathrm{rest}}$ at $\bar{F}_z = 1$ and falls monotonically with $\bar{F}_z$.
Every axle slope quoted in this paper is therefore taken at $\bar{F}_z = 1$ and at the static axle loads listed here (the geometric split $l_r/L, l_f/L$ of the total wheel load; see Section A, A2, for its 1.4 % disagreement with the measured load split of Table I).
The peak angle follows from $K = 3$ in (8), i.e. $\tan\alpha_{\mathrm{peak}} = 3\mu/\big(k_y S_1(3\bar{F}_z/k_x)\big)$, and therefore moves with μ: 12.0° at the effective $\mu = 1.05$.

Two properties of (8) shape everything downstream. It saturates at μF_z, first reached at $K = 3$, and it *never falls off* past that peak—so a Magic Formula, whose E and post-peak decay exist precisely to describe a falling branch, can match it up to the peak and not beyond. And its scale is set entirely by μ and C_{lat}, both of which the simulator publishes, so the curve of Table II is read off the plant rather than fitted to it.

That the closed form is the right one was checked rather than assumed. On 1804 ground-truth wheel samples from a live lap ($P_{99}(|\alpha|) = 3.66\,°$), each predicted from its own measured α, F_z and μ, (8) returns $R^2 = 0.99$ with 16.1 N RMSE on the front axle and 14.7 N on the rear, at biases of +2.0 N and +0.8 N. Section VI-D uses this curve to separate the error of the identified fit from the error of representing (8) in the form (4) that the controller requires.

D. CARLA_ASU_BRIDGE: interfaces and telemetry

CARLA_ASU_BRIDGE [39] is a C++ ROS 2 lifecycle node linked directly against CARLA's client library (Fig. 1). It spawns and configures the ego vehicle, drives its inputs, and republishes simulator state as standard ROS 2 topics; all identification traffic crosses this interface.

The command path is a single `ackermann_msgs/AckermannDriveStamped` on `/drive`, forwarded to CARLA's native Ackermann controller; no client-side steering PID is present, so the achieved/commanded ratio of Table I is a property of the vehicle rather than of a tuning choice in the bridge. On the feedback path, one property is specific to this architecture and is what the rest of this paper rests on: the simulator's per-wheel slip angle, normal load and lateral force are republished as ROS 2 topics, on the same simulation clock as the odometry the identifier consumes, so an identified tire model can be scored against the forces the plant actually generated rather than against a second model. Appendix B lists the full

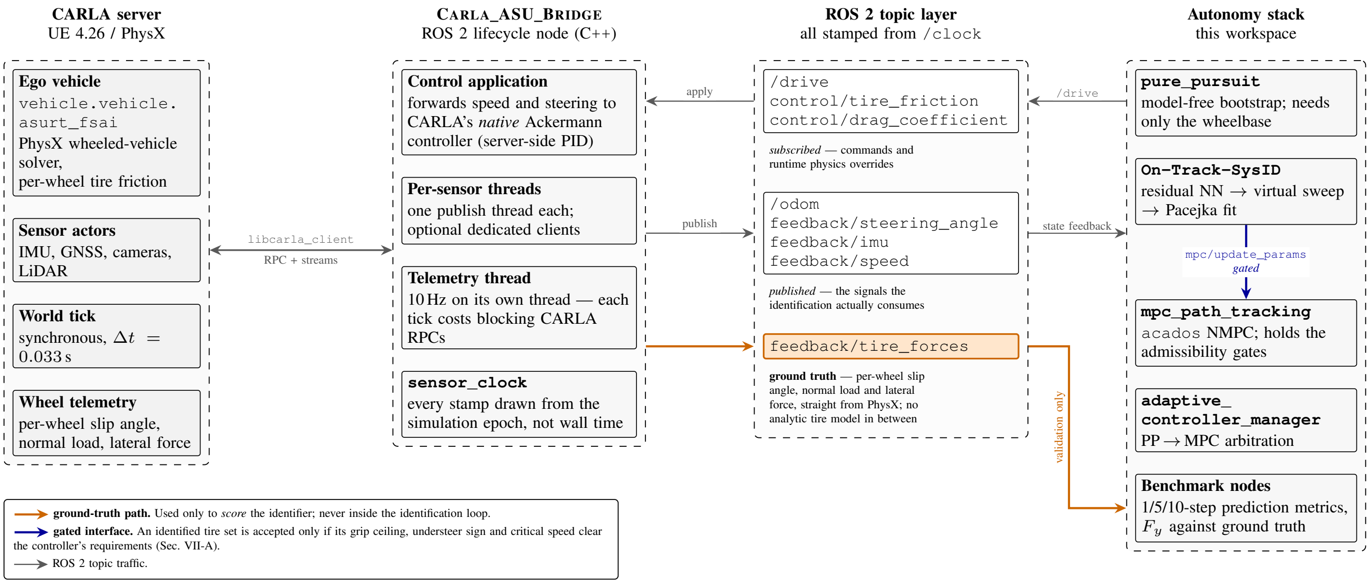


Fig. 1. Architecture of the CARLA_ASU_BRIDGE ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel telemetry on ROS 2 topics stamped from the simulation clock. The `feedback/tire_forces` path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

interface, the body-frame and sign conventions applied at the simulator–ROS 2 boundary, and the six extensions made to the bridge for this study.

E. Data acquisition and preprocessing

The vehicle is driven around the reference raceline by a model-free pure-pursuit controller, which requires no tire knowledge—only the wheelbase—and is therefore a valid bootstrap, as in the baseline [6], [3]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer by a 50 Hz timer for 30 s, gated on $v_x > 1$ m/s. The timer runs on the simulation clock, which advances only at the server's 0.033 s step, so the buffer cannot acquire a new state faster than the physics produces one and the achieved acquisition rate is the step rate, not the timer rate.

Two distinct mismatches follow from that arrangement, and they carry different consequences. The first is between the acquisition timer and the physics step: the timer asks for a sample every 0.020 s, 65 % more often than the server produces one, so a fraction of consecutive buffer entries are repeated states and the one-step residual target $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ carries a zero-order-hold artefact on those pairs. The second is between the physics step and the interval the one-step model of Section III-A is discretised at, which is set separately at $T_s = 0.035$ s (`nn_params.yaml:sample_dt`) and is 6 % longer than the step. T_s is an empirical setting: it is neither the 0.033 s step nor the 0.020 s timer period, and it is carried because the identification proved insensitive to it over the range tested, not because it matches either clock. That insensitivity was measured rather than assumed—the identification was repeated at both settings. The same two configurations identified at $T_s = 0.020$ s and at $T_s = 0.035$ s return the *same* last accepted coefficient set to four decimal places in the corrected configuration ($B, C, D, E = 7.076, 1.346, 1.009, -2.000$ front and $7.873, 1.383, 1.002, -1.024$ rear), and in the inherited configuration the coefficients move (B_f 4.72→4.80, E_f -2.44→-2.26) while D stays on its 2.0 bound on both axles at both settings. The identified tire is therefore insensitive to the mismatch, and the box-edge signature is not an artefact of it. Results in Section VI are reported at $T_s = 0.035$ s. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \to (-v_y, -\omega, -\delta)$ at unchanged v_x, which doubles the data and removes the cornering bias of a circuit. Re-identification retriggers every 30 s by default.

F. How the architecture closes the loop

The four stages of the identification pipeline map onto the architecture as follows. *Acquisition* uses `/odom`, `feedback/steering_angle` and `/drive`, all stamped on the simulation clock, so that the interval the one-step model is discretised at can be matched to the interval the physics actually advanced (Section IV-E) rather than to the acquisition timer. *Preprocessing* and *fitting* run inside the identification node against that buffer. *Validation* draws on `feedback/tire_forces` and `feedback/imu`, which never enter the identification loop, so the estimator is scored against a channel it did not observe. *Deployment* pushes the fitted coefficients over a ROS 2 service to the controller behind the gates of Section VII. Only the last stage leaves the architecture, which is what lets Section VI attribute a closed-loop difference to the identified model rather than to the interface that carries it.

V. IDENTIFICATION METHODOLOGY

This section describes the identification pipeline. Section V-A covers the residual model, V-B–V-C the two changes

in: buffer $\mathcal{D} = \{v_x, v_y, \omega, \delta^{\text{ach}}\}_{1:K}$ (30 s), prior Φ_p^0, bounds $[\Phi_p^-, \Phi_p^+]$
out: Φ_p, or UNIDENTIFIED

$\delta \leftarrow$ achieved road-wheel angle, command on staleness ▷
$m, l_f, l_r \leftarrow$ measured static wheel loads ▷
if first cycle **then** ▷
  $\hat{\mu}_f, \hat{\mu}_r \leftarrow$ BRUSHFIT($\mathcal{D}$); release only if gated
  $D_a^0 \leftarrow \max(D_a^0, \hat{\mu}_a)$ (floor only, never a replacement)
$\Phi_p^{\text{tik}} \leftarrow \Phi_p^0$ (frozen for all iterations) ▷
$v \leftarrow \text{clip}(\overline{v_x}(\mathcal{D}), v_{\min}, \infty)$ ▷
$\delta_{\text{sweep}} \leftarrow \text{clip}(1.5\, P_{99}(|\delta|), \delta_{\min}, \delta_{\max})$ ▷
$\mu_{\text{reach}} \leftarrow v^2 \delta_{\text{sweep}}/(gL)$; **warn if** $\mu_{\text{reach}} < \max_a D_a^0$ ▷
$\Phi_p \leftarrow \Phi_p^0$
for $i = 1$ **to** N_{it} **do**
  $f_{NN} \leftarrow$ TRAINRESIDUAL($\mathcal{D}, \Phi_p$) (re-initialised each i)
  $\mathcal{S} \leftarrow$ ROLLOUT($\Phi_p, f_{NN}, v, \delta_{\text{sweep}}$)
  $\Phi_p \leftarrow \arg\min \|F_y(\Phi_p; \mathcal{S}) - \hat{F}_y(\mathcal{S})\|^2 + \lambda \|\Phi_p - \Phi_p^{\text{tik}}\|^2$
    subject to $\Phi_p \in [\Phi_p^-, \Phi_p^+]$, started from Φ_p of $i-1$
if any Φ_p component on a bound **then** report the degeneracy ▷
if not GATES(Φ_p) **then return** UNIDENTIFIED ▷
submit Φ_p; keep the previous model until acknowledged ▷

: GATES: axle stiffness sign and magnitude; $l_f C_f < l_r C_r$ and v_{crit} outside the speed range; grip ceiling $\geq$ the trajectory's lateral demand (Section VII).

Algorithm 1: Corrected on-track identification cycle. ▷ marks a change to the inherited pipeline [6].

to the virtual-data step that constitute the paper's central methodological contribution, V-D the fit itself, V-E the measurement chain, and V-G the validation protocol. Figure 2 summarises the loop and Algorithm 1 states it in the form needed to reimplement it; the lines marked ▷ are this paper's changes to the inherited pipeline.

A. Residual model

The residual carries whatever the nominal model (1)–(4) does not represent, and how that map is parameterised is a modelling decision rather than a hyperparameter. Two decisions are made here, both of the same kind: known vehicle physics is used to shape the residual's *input space* and, optionally, its *objective* (Section V-A1), so that the network's small capacity is spent on what is genuinely unmodelled instead of on rediscovering structure that (3) and (1)–(2) already state exactly. The rest of this subsection states both and why they were chosen; both are active in the configuration evaluated in Section VI, and neither is scored on its own.

The default residual is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU, 58 parameters, full-batch Adam [40] at 5×10^{-4} on an MSE objective, in PyTorch [41]. Because the full-scale residual has different structure from the scaled platform's, the architecture is selectable: `baseline`, `wide` (16 hidden units, 114 parameters, weight decay), `physics_inputs` (74 parameters), `ensemble` (three averaged members, 174 parameters), and `s4` (102–110 parameters). The results below use the baseline MSE objective, so that the identifiability statements of Sections V-B–V-C are not confounded with a change of residual.

1) Physics-augmented inputs and objective: Two options shape the residual with known vehicle physics rather than with capacity. The *input space* is augmented with the axle slip angles of (3): the tire law (4) is a function of (α, F_z) and of nothing else in $\mathbf{z}$, so supplying α_f, α_r directly spares the network from rediscovering a relation the model already states exactly. The *objective*

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_s \mathcal{L}_{\text{steady}} + \lambda_m \mathcal{L}_{\text{sym}} + \lambda_g \mathcal{L}_{\text{smooth}}, \quad (9)$$

with $\lambda_s = 0.1$, $\lambda_m = 0.05$, $\lambda_g = 10^{-4}$, adds three penalties to the one-step MSE [42], [21]. Each answers a weakness of training a residual on one lap of ordinary driving. $\mathcal{L}_{\text{steady}}$ keeps the two residual channels jointly consistent with (1)–(2) on a quasi-steady mask, so they cannot imply an axle force pair no tire could produce; it is that force pair, not the state prediction, that stage 3 fits. $\mathcal{L}_{\text{sym}}$ enforces the odd symmetry $e(-\mathbf{z}) = -e(\mathbf{z})$ that a laterally symmetric vehicle has, supplying from geometry the half of the input domain a circuit under-covers. $\mathcal{L}_{\text{smooth}}$ is a soft Lipschitz bound that decides how fast the residual may grow inside the range the extrapolation bound of Section V-C allows it to be queried over. The derivations, the masks and the precomputation that makes the nominal-model half of (9) free are in the supplementary material.

2) Temporal residual: The `s4` architecture replaces the memoryless MLP with an S4D diagonal state-space residual [23] over a sliding window of $W = 20$ samples (0.7 s at $T_s = 0.035$ s), so the residual can carry the tire-relaxation and yaw-damping dynamics that a per-sample map cannot. The architecture is not claimed as a contribution—it is also in concurrent work [22]—and neither is the 8.7× reduction in its training cost, which is an exact algebraic rewrite of the same forward pass and changes no identified coefficient. Both are specified, derived and measured in the supplementary material; nothing in the argument of this paper depends on them, and the identifiability result of Section V-B holds for either residual architecture.

B. Excitation-aware virtual rollout

This is the central correction. Consider stage 2 of Section III-D: a constant-speed rollout at v with the steering ramped to δ_{sweep}. In quasi-steady cornering the lateral acceleration a single-track vehicle develops kinematically is $a_y = v^2 \kappa$ with $\kappa \approx \delta/L$, so the largest friction coefficient the rollout can *demand* of the tire is

$$\boxed{\mu_{\text{reach}} \approx \frac{v^2\, \delta_{\text{sweep}}}{g\, L}} \quad (10)$$

Equation (10) contains no tire parameter: it is a property of the rollout configuration alone. It is also an *upper* bound on what the rollout demands rather than an equality, because $\kappa \approx \delta/L$ is the Ackermann (kinematic) curvature; the steady-state curvature a compliant vehicle actually holds is $\kappa = \delta/(L + K_{us} v^2/g)$, which is smaller. Reading (10) as a ceiling is what the argument needs: if even the ceiling sits below D, the peak is certainly not reached. If $\mu_{\text{reach}} < D$, the synthetic data never leaves the linear ramp of (4), only the product BCD is identifiable (Section III-C), and the bounded optimiser resolves the residual freedom on a box edge—the signature measured in Section VI-D.

The baseline implementation selects the rollout speed as $v = \text{clip}(\overline{v_x}, 2, 4)$ m/s, a value representative of a 1:10 platform.

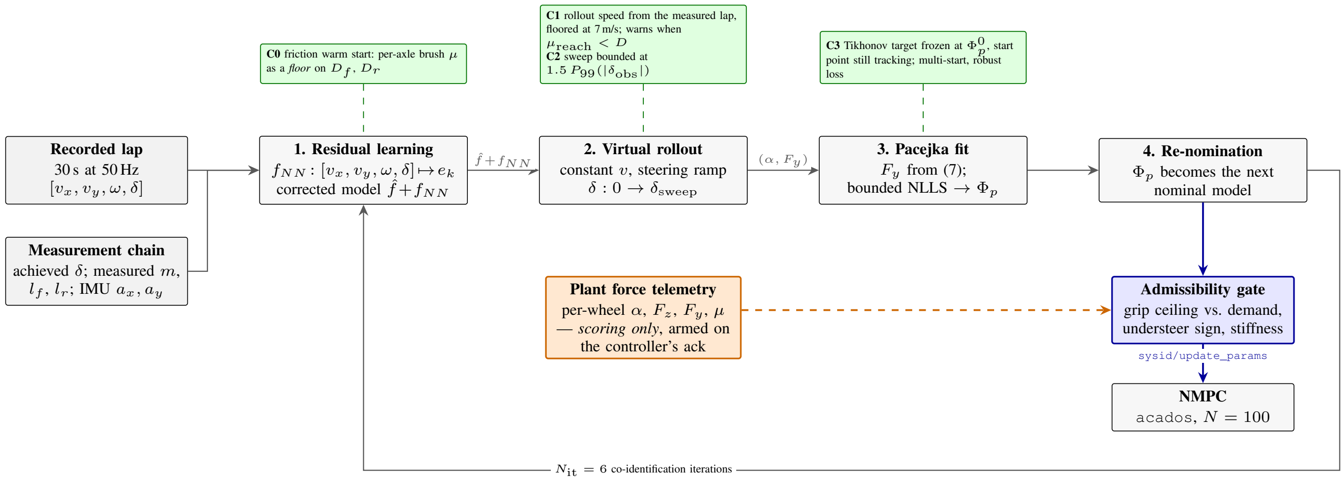


Fig. 2. The identification pipeline. Recorded lap data trains the residual network; the corrected model is rolled out through a synthetic steering sweep; Pacejka coefficients are fitted to that sweep; the fit becomes the next nominal model. Green marks this paper's changes to the inherited loop: the friction warm start applied as a floor on D (Section V-E3), the excitation-aware rollout speed and the extrapolation-bounded sweep (Sections V-B–V-C), and the frozen regularisation target of (11). Blue is the gated model interface to the controller (Section VII); orange is the plant's own per-wheel force telemetry, which scores the result, is armed only once the controller has acknowledged the submitted model (Section V-F), and is never admitted into the loop.

On our vehicle, with $\delta_{\text{sweep}} = 0.4$ rad and $L = 1.5334$ m, (10) gives $\mu_{\text{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than 0.43 g whatever the tire is. At 1:10 scale, with $L \approx 0.32$ m, the same rollout speed yields $\mu_{\text{reach}} \approx 2.0$ and the pathology is invisible.

1) Correction: The rollout speed is now taken from the speed the vehicle was actually driven at, clamped by configuration bounds (lower bound 15 m/s, no upper bound), and the pipeline warns whenever (10) evaluates below the prior's D—turning a silent degeneracy into a startup diagnostic. The same bound is the right tool for choosing a data-collection speed: identification data should be gathered at $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$.

C. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure. The rollout queries the residual network at steering angles it has never seen: on a pure-pursuit lap the observed $|\delta|$ stays below ≈ 0.06 rad while the sweep ran to 0.4 rad. The fit then reads the network's extrapolation as additional grip, and because each iteration's answer became both the next iteration's nominal model *and* its regularisation target, the error compounds: D walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed 1.7 g on a vehicle measuring 1.0 g in steady state.

Two changes close the loop:

1) **Extrapolation bound.** The sweep terminates at $\delta_{\text{sweep}} = \text{clip}\big(\lambda_e \cdot P_{99}(|\delta_{\text{obs}}|),\, \delta_{\min},\, \delta_{\max}\big)$ with $\lambda_e = 1.5$, $\delta_{\min} = 0.05$ rad, $\delta_{\max} = 0.54$ rad, so the network is never queried more than 50 % beyond the steering range it was trained on. Both $\delta_{\max}$ and the inherited $\delta_{\text{sweep}} = 0.4$ rad lie above the plant's 16.0° (0.279 rad) mechanical lock—$\delta_{\max}$ by a factor of 1.9, the inherited sweep by 43 %. That is deliberate: the sweep is a *virtual* rollout of the corrected model, and clamping it to the lock would cap μ_{reach} in (10) for a second time—by the actuator instead of by the rollout speed—which is the failure this section exists to remove. The binding limit in practice is the data-driven $\lambda_e P_{99}(|\delta_{\text{obs}}|)$ term, not $\delta_{\max}$; the residual is nonetheless evaluated outside the plant's reachable steering set, which is the extrapolation recorded in Section A, A6.
2) **Frozen prior.** The Tikhonov target is frozen at the configured prior Φ_p^0 for all iterations, while the optimiser's *start point* continues to track the previous iteration. The per-axle objective is

$$\min_{\phi} \sum_k \rho\big(r_k(\phi)\big) \;+\; \sum_j w_j \big(\phi_j - \phi_j^0\big)^2, \qquad (11)$$

with r_k the force residual, ρ a soft-ℓ_1 robust loss and w_j scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The regulariser exists specifically to hold the coefficients the manoeuvre does not excite—E above all—at their nominal value instead of on a box edge.

D. Pacejka fit

Fitting applies the steady-state force recovery of (7) to the synthetic data under box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle. Relative to the baseline we (i) raised the number of jittered multi-starts from 1 to 8, (ii) added a derivative-free simplex [43] and a global differential-evolution solver [44] as alternatives to trust-region reflective [45], all through the same optimisation library [46]; population-based global search on Magic-Formula coefficients follows Cabrera *et al.* [47]. The prior carries the measured tire of Table II. One caveat on (i): `num_starts` is honoured by the trust-region and simplex solvers only—differential evolution is already a population search and ignores it—so the multi-start claim describes the `Baseline-NN-MSE` arm and not the shipped `Ours` arm.

Neither (i) nor (ii) addresses the degeneracy, and both are verified as secondary in Section VI-B, which sweeps rollout speed against true D on a known tire: with the rollout speed left at 4 m/s every tire tested fits back as $\hat{D} \approx 0.5$ with coefficients railed, and the answer does not move with the multi-start seed or with a jitter seven times wider.

E. Measurement-side corrections

Three corrections apply to the measurement chain rather than to the estimator.

1) Achieved versus commanded steering: α_f in (3) was built from the `/drive` setpoint. The measured achieved/commanded ratio is 0.76, so α_f was systematically $\approx 25\,\%$ too large and the fitted front stiffness $\approx 20\,\%$ too small (13725 N/rad against 17198 N/rad fitted to the simulator's own per-wheel forces over the same run). Using `feedback/steering_angle` recovers 18486 N/rad. All three are effective slopes fitted over that run's load range and are therefore below the $\bar{F}_z = 1$ closed-form value of Table II; the comparison is between them and not against the tabulated slope.

2) Mass and geometry: m and l_{wb} are taken from the measured static wheel loads (Table I) rather than from the configuration file, and $l_{wb} \equiv l_f + l_r$ is enforced by a regression test, because (7) uses the wheelbase for the normal loads and the axle split for the force division; the two disagreeing is a silent bias.

3) Proprioceptive friction warm-start: The third correction is to where the peak factor's prior comes from. The inherited pipeline had none: D started from a configuration constant and was left to the fit. What follows recovers a prior for it from the sensors the vehicle already carries. Four steps: axle forces and slip angles from the body's own motion (a); a two-parameter brush fit that reads μ from the *curvature* of those forces rather than from their asymptote (b); the release gate that says when that reading means anything (c); and the rule that the released value may only ever raise D (d).

a) Axle forces and slip angles without a force sensor: Newton–Euler on the single-track body of (1)–(2), solved for the axle forces rather than for the accelerations, gives

$$F_{y,f} = \frac{I_z\dot{\omega} + l_r m a_y}{L\cos\delta}, \qquad F_{y,r} = \frac{l_f m a_y - I_z\dot{\omega}}{L}, \tag{12}$$

which requires the IMU's lateral acceleration and the yaw acceleration and no tire model at all. Vertical loads carry the longitudinal transfer $F_{z,f/r} = m(g\,l_{r/f} \mp a_x h_{cg})/L$ [34]. The yaw acceleration is taken as the derivative of a local polynomial fit to the yaw rate (Savitzky–Golay, window 21, order 2) [48] rather than by differencing, because $I_z\dot{\omega}$ is a $\approx 10\,\%$ term in $F_{y,f}$ on this vehicle and a differenced odometry yaw rate puts its noise straight into it. Slip angles come from (3) on the buffer that is already collected. Where no IMU is available the accelerations are differentiated from the states by the same filter, and where one is available the RMS difference between the two is logged as a bias diagnostic.

b) Joint fit of stiffness and peak friction: Per axle, both the cornering stiffness and the peak friction are fitted to the Fiala brush model [25], [49], [4]

$$F_y = \mu F_z\big(1 - (1-\xi)^3\big)\operatorname{sgn}\alpha, \quad \xi = \min\Big(\frac{C_\alpha|\tan\alpha|}{3\mu F_z}, 1\Big), \tag{13}$$

by bounded nonlinear least squares with a soft-ℓ_1 loss [46], [45]. The brush model is used here in place of (4) for one reason: it has two parameters, both of which the manoeuvre can support, whereas (4) has four and is exactly the model whose non-identifiability at low slip this paper is about.

What makes this beat the utilisation bound is that μ enters (13) through the *curvature* of $F_y(\alpha)$ and not only through its asymptote. Writing $r = |F_y|/(C_\alpha|\tan\alpha|)$ for the force's departure from its own linear extrapolation, (13) inverts in closed form to

$$\mu = \frac{2k}{3 - \sqrt{3(4r-1)}}, \qquad k = \frac{C_\alpha|\tan\alpha|}{3F_z}, \tag{14}$$

so a 30 % departure from linearity ($r = 0.7$) already pins μ, well below the limit. Equation (14) is used only for the excitation diagnostics; C_α and μ are fitted jointly, because a two-stage fit would inherit the bias of estimating C_α on data that is itself already slightly nonlinear.

c) The gate is the feature: Equation (14) also states its own failure mode: $\mathrm{d}\mu/\mathrm{d}r$ grows without bound as $r \to 1$, i.e. as the tire curve stays linear. The estimate is therefore released only when three conditions hold—peak friction utilisation $\max|F_y|/(\hat{\mu}F_z) \geq 0.40$, relative precision $\sigma_{\hat{\mu}}/\hat{\mu} \leq 0.15$ from the Gauss–Newton covariance, and $\hat{\mu}$ not resting on its own search bound—and otherwise the module logs which condition failed and falls back to the utilisation bound.

The μ in the utilisation test is the *fitted* $\hat{\mu}$ of the same least-squares call, not an independent one, so that test is on its own circular: a fit that underestimates μ inflates its own utilisation and passes itself. It is therefore never used alone. The two companion conditions break the circularity: a fit driven low by data that is still linear returns a wide $\sigma_{\hat{\mu}}$, and a fit driven onto its search bound is rejected as railed. Whatever still slips through is made one-sided by the floor of Section V-E3d, since an underestimate cannot lower D. The 40 % floor is not tuned here: it is the excitation level the lateral-dynamics friction-estimation literature converges on, as surveyed by Acosta *et al.* [16], whose review reports estimators failing below $|a_y| = 0.4\,g$ "due to the proximity of the tyre force curves in the low slip regions" and a 40–80 % utilisation working range. Samples with $|a_x| > 0.35\,g$ are dropped rather than corrected, since the axle longitudinal force split a friction-ellipse correction would need is not observable from this sensor set; samples below 2.5 m/s are dropped because (3) is ill-conditioned there; and samples whose force opposes their own slip angle are dropped as sign errors or noise.

d) Applied as a floor, unconditionally: The per-axle μ replaces the scalar bound as the initial guess for D_f and D_r where it is released, but—for either input, and however confident the fit—it can only ever *raise* D, never lower it. This is not conservatism for its own sake; two mechanisms depend on it. First, the frozen Tikhonov target of (11) is taken from the warm-started prior immediately after the warm start is applied, so whatever lands there is both the initial guess and the regularisation target of all six iterations. Second, the controller re-ingests the reference speed profile at $\min(a^{\mathrm{cfg}}_{y,\max},\, a^{\max}_y\eta)$ on every accepted model update, so a lowered D rewrites the reference under the running controller; a lowered D that also *moves* between cycles rewrites it repeatedly. The floor makes the estimate a one-way influence on the reference, which is the property the interface needs.

The gate cannot substitute for the floor, and it is worth being precise about why: σ_μ/μ from the fit covariance is a *precision* figure, not an accuracy one. A wrong mass, a wrong yaw inertia, an IMU bias or a road bank all fit tightly and wrongly through (12), since axle force scales linearly with m. On this vehicle the configuration file's 240.0 kg against the 269.6 kg the wheels actually carry biased μ 11 % low on its own—the 29.6 kg shortfall taken on the 269.6 kg the wheels carry—until the measured value replaced it (Table I); that is measurement-chain work (Section V-E) and not something a confidence interval can see. The same holds for the friction the result is compared against: the configured 1.5 is not the 1.05 the physics step uses, and an estimate validated against the former would be judged 30 % short $(0.45/1.5)$ while being correct. Of the inputs to (12), only I_z now remains unmeasured on this vehicle; it enters $F_{y,f}$ at the 10 % level and is listed among the assumptions of Section A.

F. Deployment

The identifier runs as a single ROS 2 node whose acquisition and state-machine timer ticks at 50 Hz (the 20 Hz quoted elsewhere in this paper is the controller's period, not the identifier's), with a rolling FIFO buffer of the collected signals, cycling through four states: collect a full buffer, train, await the controller's acknowledgement of the submitted coefficients, then run—re-identifying every $T_{\text{reid}} = 30$ s from the newest buffer. Three interface details matter for reproducing the results. The achieved road-wheel angle is used when the platform publishes one and falls back to the commanded setpoint on staleness, with the substitution logged (Section V-E). The friction warm start is applied on the *first* identification cycle only; every later re-identification cold-starts D from the static `pacejka_model` prior of configuration. That restriction is the safety property, not an implementation shortcut: the warm start is both the solver's start point and, through the frozen prior of (11), its regularisation target, so recomputing it each cycle would rewrite the tire model—and through it the MPC's reference speed profile—under the running controller. Finally, the identified coefficients are submitted over a service call and the node keeps the previous model if the call is not acknowledged, so a rejected identification degrades to the last accepted model rather than to no model; the peak-friction outputs are published on a latched diagnostic topic so a late-starting consumer still sees the current value.

That acknowledgement is also what arms the scoring. The node's own online metrics, and the parameter forward to the passive benchmark node, both fire from the acknowledgement handler rather than from the end of training. The distinction is not cosmetic: a submission the controller *rejects* on its admissibility gate (Section VII) is a model the car never drove on, and the seconds between a fit completing and its acknowledgement landing are seconds in which the car is still driving the previous model. Scoring either of them attributes to the identified model behaviour that belongs to another one, which is exactly the confusion the validation protocol below exists to avoid.

G. Validation protocol

Three levels of validation are used, in increasing strength:

1) **One-step prediction** on a held-out run, as in the baseline: RMSE and R^2 of v_y and ω under $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. A dedicated ROS 2 node extends this to configurable multi-step horizons (1/5/10 steps) with Euler/Heun/RK4 integration and online statistics in Welford's form [50]. Samples are admitted only above 1 m/s, the same gate the identification data itself uses. This is not a division-by-zero guard: since $\alpha = \arctan(v_y/v_x)$, a few tenths of a metre per second of lateral drift saturates the tire curve and the one-step model predicts yaw rates of several rad/s against a ground truth of a few tenths, so a looser gate lets the low-speed bootstrap dominate a metric that is meant to describe the racing regime the model was fitted on.
2) **Direct force comparison** against `feedback/tire_forces`: the identified curve is evaluated at the measured (α, F_z) and compared per axle against the measured F_y, and against the plant's own curve (8) evaluated at the same points (Table II), which separates the error of the fit from the error of representing a PhysX tire.
3) **Closed-loop tracking** under the MPC using the identified model: mean and maximum lateral deviation, its time-weighted integral, heading error and solver cost, split by the active controller so that the pure-pursuit acquisition phase is not scored against a model it does not use. Lap counting runs from the first crossing of the raceline's $s = 0$, so the drive from the spawn point up to the start line is an untimed out-lap (Section VI-H).

All three levels score the *accepted* model only, for the reason given in Section V-F.

VI. Experiments and Results

A. Setup

All experiments run on the architecture of Section IV: the Formula AI vehicle in CARLA [8] driven through ROS 2 Humble, with the server in synchronous mode at a 0.033 s fixed physics step. The identifier collects 30 s of data at 50 Hz and re-identifies from the newest buffer, matching the baseline's data budget [6]. Ground truth for forces, slip angles and normal loads is the simulator's own per-wheel telemetry, which the identifier never reads.

1) Surface: Every run reported here uses a *static* friction surface: the per-wheel coefficient is held at its configured value of 1.5 for the whole run, and the road-surface factor of 0.70 that PhysX applies on top of it makes the effective coefficient the identifier is scored against 1.05, constant to the last digit the simulator reports over all 8186 and 9355 force-validation samples of the two runs. Time-varying friction is out of scope here (Section IX).

TABLE III
THE TWO CONFIGURATIONS.

Setting	`Baseline-NN-MSE`	`Ours`
Residual architecture	MLP (`baseline`)	S4D (`s4`)
Residual objective	One-step MSE	Physics-informed, (9)
Pacejka solver	Trust-region [45]	Differential evolution [44]
Friction warm start	Off	On, floor only

Everything not listed is identical between the two, including the excitation-aware rollout of Section V-B, which both use.

2) Reference trajectory: Both runs follow the same optimised raceline (Fig. 3), produced with the TUM global race trajectory optimisation package [51] under its minimum-curvature formulation [33] on a recorded centreline of the CARLA silverstone circuit: a closed 5828.2 m lap of 2916 waypoints at 2.0 m spacing, reference speed 9.62–25.78 m/s, minimum corner radius 18.7 m ($|\kappa|_{\max} = 0.0536\,/\mathrm{m}$). Its peak lateral demand is $\max(v_x^2|\kappa|) = 4.98\,\mathrm{m/s}^2$ ($0.51\,g$), which saturates the flat $5.0\,\mathrm{m/s}^2$ `ggv` envelope the optimiser was given and stays at roughly half the $1.0\,g$ the plant sustains. This trajectory replaces the earlier one, whose $1.22\,g$ demand the admissibility gate of Section VII rejected against the same plant; the regeneration, not a relaxed gate, is what unblocked the closed-loop runs below.

3) The two configurations: Two configurations of the same pipeline are run back to back on that trajectory, three timed laps each, with an untimed out-lap from the spawn point to the raceline's $s = 0$. They differ in the four settings of Table III and in nothing else: the vehicle, the map, the trajectory, the controller, the data budget, the co-identification loop count and the Pacejka box constraints are shared. `Baseline-NN-MSE` is the inherited configuration—MLP residual, one-step MSE objective, local trust-region fit, no friction warm start—evaluated at full scale on the corrected rollout, so the comparison isolates the four settings rather than reproducing the published pipeline in full (that comparison is Section VI-I). `Ours` is the shipped configuration. Because the four settings move together, the comparison below is a configuration comparison and not a per-factor ablation. It is also not a test of this paper's central contribution: the excitation-aware rollout of Section V-B is active in *both* arms, so what Section VI-H measures is the four secondary settings on top of that correction. The correction itself is evidenced in Section VI-B.

4) Provenance of the tire prior: The prior Φ_p^0 that both arms start from, and that the Tikhonov term of (11) pulls toward, is the measured tire of Table II. It was obtained by fitting (4) to CARLA's own per-wheel telemetry (`feedback/tire_forces`: slip angle, lateral force, normal load) over a dedicated limit-excitation run that reached $\alpha = 0.37\,\mathrm{rad}$, axle $\mu = 1.05$ and $1.50\,g$, by differential evolution, at 135 N front and 85 N rear residual RMSE. Two consequences have to be stated plainly. First, that is the same channel the force and friction results below are scored against, so the prior is not independent of the validation data. Second, the manoeuvre that produced it is one the pipeline itself is not allowed to run: the raceline of Section VI-A2 peaks at $0.51\,g$ and never approaches the tire's peak, which is precisely the excitation deficit this paper is about.

The consequence for the numbers is direct. On all six accepted identifications the `Ours` arm returns the eight prior coefficients to four decimal places (Section VI-D), so the peak-friction accuracy reported in Section VI-E—μ RMSE 0.041 front and 0.048 rear—is the accuracy of a prior that the fit did not move, not a peak factor recovered from the lap. What the corrected pipeline demonstrably does on this trajectory is *preserve* a physically valid peak factor where the inherited configuration destroys it by railing D on its bound; that, and not identification of the peak, is the claim these runs support. Section VI-B is where the peak factor is shown to be recoverable from data, on a plant where the excitation can be varied.

5) What is measured: Three families of quantity are recorded, each against the plant's own telemetry: the identified tire model (coefficients, axle force, peak friction), the one-step state prediction, and the closed-loop tracking of the MPC that consumes the model. Every figure in this section is exported together with the CSV holding the samples behind it, and the cross-run comparison reads those CSVs rather than re-deriving any metric, so a figure and the table cannot disagree.

B. Identifiability of the peak factor against rollout excitation

The identifiability claim of Section V-B is a claim about the rollout configuration alone, so it is tested where the answer is known exactly rather than in CARLA. A tire with a known peak factor D is rolled out through the synthetic steering sweep of stage 2—the shipped rollout, with the residual network set to zero—and fitted back with the shipped solver. The fit is run *unregularised* and started from the true coefficients, so nothing the fit loses is lost to the prior, to the start point or to a local minimum; everything it loses is lost to excitation. Table IV and Fig. 4 report the grid.

Three axes are swept—rollout speed v, sweep amplitude δ_{sweep}, and true D—and each cell is repeated over five noise realisations. The two configuration axes are swept one at a time, because μ_{reach} is a function of both and a claim that it, and not the speed, is what governs the fit cannot be tested by moving the speed alone. The repetition is over noise and not over the solver because the noise-free fit is exactly seed-invariant here: five multi-start seeds return the same coefficients to four decimal places, and so does a jitter width seven times wider, which is the direct measurement behind the statement that the degeneracy is not a solver problem. The injected noise is Gaussian on the rolled-out states at the measured one-step persistence RMSE of the two CARLA runs, $\sigma_{v_y} = 0.009\,\mathrm{m/s}$ and $\sigma_\omega = 0.005\,\mathrm{rad/s}$ (Table V), and is what the intervals in Table IV are computed over. Because the front and rear fits of one rollout share that rollout, the interval is taken over the five realisations and not over the ten fits.

The grid says three things. First, the inherited 4 m/s rollout is uninformative about D at *every* grip level tested: a 0.70, a 1.05, a 1.40 and a 1.80 tire all come back as $\hat{D}$ between 0.50 and 0.59, i.e. the fit reports μ_{reach} and not the tire, with two to seven of the eight coefficients on a bound. Second, the boundary is $\mu_{\mathrm{reach}} \gtrsim D$ and not a speed: at 6 m/s,

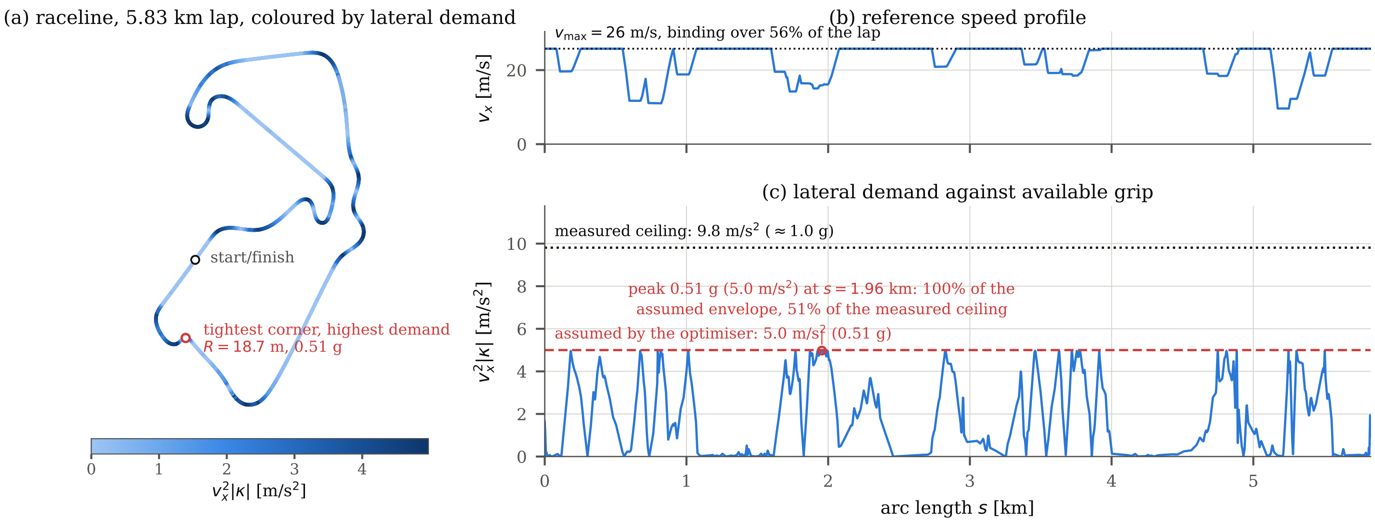


Fig. 3. The optimised raceline used for both runs, generated with the TUM global race trajectory optimisation package [51] under its minimum-curvature formulation [33] from a recorded centreline of the CARLA silverstone circuit. (a) Track layout, coloured by lateral demand; the marked corner ($R = 18.7$ m) is the tightest on the lap. (b) Reference speed profile, 9.62–25.78 m/s. (c) Lateral demand $v_x^2|\kappa|$ along the lap against the flat 5.0 m/s^2 envelope handed to the optimiser (dashed) and the grip the vehicle has (dotted, $\approx 1.0\,g$, measured). The profile saturates the envelope and no segment exceeds the plant's grip.

TABLE IV
PEAK FACTOR $\hat{D}$ RECOVERED FROM A KNOWN TIRE, ALONG BOTH ARGUMENTS OF μ_{reach}.

swept value	μ_{reach}	$\hat{D}$ (railed of 8) at true D of 0.70	1.05	1.40	1.80
(a) rollout speed v [m/s] swept at $\delta_{\text{sweep}} = 0.40$ rad					
4	0.43	0.52±0.036 (1.8)	0.57±0.013 (2.8)	0.59±0.018 (3.8)	0.50±0.001 (7)
6	0.96	0.68±0.001 (1)	1.04±0.014 (1.2)	1.22±0.066 (1.8)	1.26±0.097 (2)
8	1.70	0.69±0.001 (1)	1.02±0.001 (0.8)	1.36±0.002 (0)	1.79±0.025 (1.2)
10	2.66	0.70±0.000 (0)	1.03±0.001 (0.4)	1.37±0.002 (1)	1.75±0.002 (0)
13	4.49	0.72±0.001 (0)	1.05±0.002 (0)	1.38±0.001 (0.4)	1.77±0.002 (0.4)
15	5.98	0.76±0.009 (1)	1.05±0.002 (0)	1.39±0.002 (0)	1.78±0.001 (0)
20	10.64	0.92±0.004 (1)	1.14±0.005 (0.4)	1.43±0.002 (0)	1.80±0.005 (0)
(b) δ_{sweep} [rad] swept at $v = 15$ m/s					
0.03	0.45	0.80±0.019 (2)	1.10±0.041 (2)	1.08±0.182 (2)	1.09±0.397 (2)
0.06	0.90	0.92±0.003 (2)	1.30±0.008 (2)	1.68±0.017 (2)	1.80±0.232 (3.4)
0.10	1.50	0.96±0.004 (2)	1.34±0.016 (2)	1.60±0.017 (1)	1.97±0.030 (1.4)
0.15	2.24	0.99±0.004 (2)	1.38±0.003 (2)	1.71±0.019 (2)	1.98±0.009 (1)
0.25	3.74	0.77±0.002 (2)	1.12±0.004 (2)	1.81±0.020 (2)	2.00±0.000 (2)
0.40	5.98	0.76±0.009 (1)	1.05±0.002 (0)	1.39±0.002 (0)	1.78±0.001 (0)
0.54	8.08	0.83±0.004 (1)	1.08±0.002 (0)	1.38±0.001 (0)	1.78±0.008 (0)

Unregularised, started from the true coefficients. Each cell is $\hat{D} \pm$ the half-width of its 95 % t-interval over the five noise realisations, with the mean number of the eight coefficients resting on a box constraint in parentheses. The two axles of one rollout are not independent, so each realisation contributes the mean of its own $(\hat{D}_f, \hat{D}_r)$ and $n = 5$; per-axle means and intervals are in the supplementary material.
μ_{reach} is (10) at $L = 1.5334$ m; the inherited configuration is the first row of (a). It is the Ackermann *upper* bound, and above $\mu_{\text{reach}} \approx 2$ it detaches from what the rollout realises: at $v = 20$ m/s it reads 10.64 while the rollout's realised $\max|v_x\omega|/g$ is 1.40–2.18 across the four tires. A large μ_{reach} means "far past saturation", not a demand the tire ever sees.

where $\mu_{\text{reach}} = 0.96$, the 0.70 and 1.05 tires are recovered to within 3 % while the 1.40 and 1.80 tires are 13 % and 30 % short. The boundary is soft rather than sharp: the 1.05 tire is recovered to 1 % at $\mu_{\text{reach}} = 0.96$, i.e. slightly *below* its own D, because a rollout that stops just short of the peak still traverses the curvature that separates B, C and D. The bound predicts where the information runs out, not a step. Above $\mu_{\text{reach}} = 1.7$ every tire is recovered to within 4 % with no systematic railing. Third, the failure is not the solver's: the noise-free fit is invariant to the multi-start seed and to the jitter width, so multi-start, a global solver or a better start point cannot recover a coefficient the data does not contain—which is why Section V-D reports the extra starts and the global solver as secondary. The one caveat is at the top of the sweep, where a rollout far past the peak begins to over-report a low-grip tire ($\hat{D} = 0.92$ for a 0.70 tire at 20 m/s), because the Magic Formula's falling branch is being fitted to data the plant's own curve does not produce; the extrapolation bound of Section V-C is what keeps the shipped configuration away from that regime.

Panel (b) of Table IV sweeps the other argument of (10): δ_{sweep} from 0.03 rad to the shipped $\delta_{\max} = 0.54$ rad, at a fixed $v = 15$ m/s. It confirms that speed is not the governing variable—starving the sweep amplitude destroys D just as

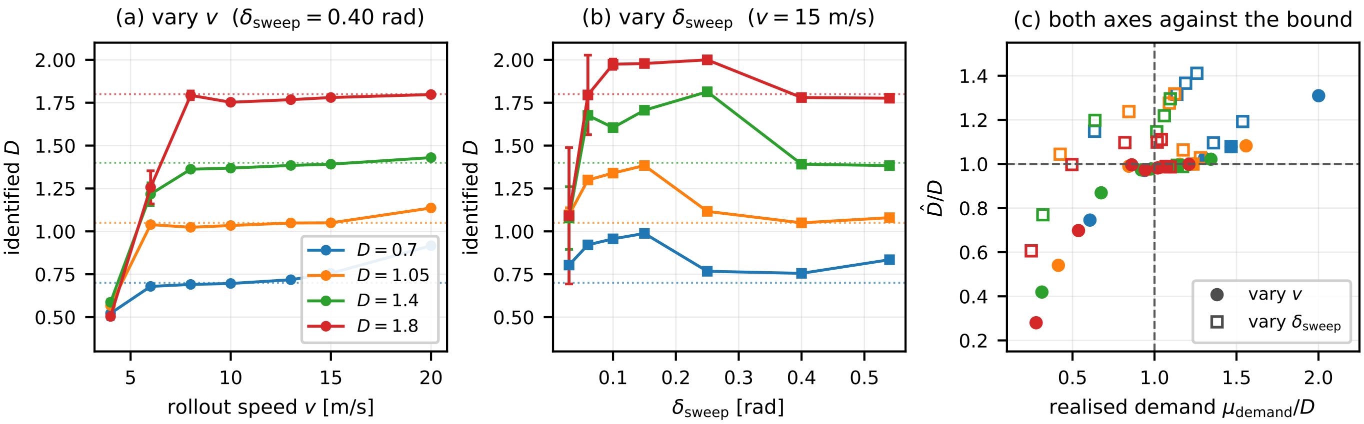


Fig. 4. The grid of Table IV, unregularised and started from the true coefficients. (a) and (b) sweep the two arguments of (10) one at a time, one series per true D (dotted lines), mean $\pm$ 95 % interval over the five noise realisations. Neither axis is a speed effect: starving δ_{sweep} at a fixed 15 m/s destroys D as thoroughly as starving v, and the 1.8 tire rails on its 2.0 bound while doing so. (c) reads both against the demand the rollout *realises*, $\max |v_x \omega|/g$, normalised by the tire's own D. Left of the dashed vertical the rollout never reaches the peak and $\hat{D}$ reports the excitation rather than the tire—low when the speed is the limiting factor (filled), high when the sweep amplitude is (open)—and the two branches close on $\hat{D}/D = 1$ from opposite sides as the realised demand passes D. The box-edge counts are the parenthesised column of Table IV.

starving the speed does, at the same fixed speed—and it corrects the shape of the claim in two ways. First, the failure is not a common bias. Where an under-speeded rollout returns *too little* grip (every tire back at $\hat{D} \approx 0.5$), an under-swept one returns too much: at $\delta_{\text{sweep}} = 0.06$ rad the 1.05 tire comes back as 1.30 and the 1.80 tire rails on the 2.0 bound, with two of the eight coefficients on a box edge in every cell. What the two axes share is that $\hat{D}$ stops being a function of the tire; the direction in which it then errs is set by the rollout, which is the signature of an unidentified parameter rather than of a biased one.

Second, the two axes do *not* collapse onto a common μ_{reach}, and the reason is the upper-bound caveat of Section V-B. Equation (10) uses the Ackermann curvature, so it tracks the realised demand only while the vehicle's own understeer is small; the realised $\max |v_x \omega|/g$ saturates near the tire's peak while μ_{reach} keeps growing, reaching 10.64 against a realised 1.40–2.18 at $v = 20$ m/s. Read against the realised demand (Fig. 4c) both axes do behave alike: recovery sets in where the rollout actually reaches the tire's peak. Equation (10) should therefore be used as it is used in the pipeline—as a cheap screening bound that is conclusive when it falls below D—and not as a measure of the excitation achieved.

One property of this experiment bounds how all of it should be read: the residual network is set to zero, so a Magic Formula is fitted to Magic-Formula data and the model structure is exactly matched. The recovery reported here is the best case, and the real pipeline's degradation at a given excitation is at least this bad.

C. Cross-run summary

Table V is the complete comparison. `Ours` is better on peak friction, on rear-axle force, on yaw rate, on lateral tracking error in every phase, on both error tails and on the worst-case QP solve time; `Baseline-NN-MSE` is better on front-axle force, on lateral velocity and on the RMS QP solve time. The rest of this section reads those rows against the figures.

D. Identified tire model

Figure 5 puts both identified curves against the plant's own curve (8) and lists the coefficients. Two things are visible.

First, no coefficient of `Ours` rests on a box constraint, in any of its six accepted identifications: $B = 7.076/7.873$ inside $[4, 20]$, $C = 1.346/1.383$ inside $[1.2, 2.2]$, $D = 1.009/1.002$ inside $[0.4, 2.0]$ and $E = -2.000/-1.024$ inside $[-3, 1]$. The front E landing on a round -2.000 is the configured prior, not a bound: E has essentially no data support here (the lap reaches 2.2° of slip, and E shapes only the region around and past the peak), which is precisely the coefficient the frozen Tikhonov target of (11) exists to hold at its nominal value. Reading it as an identified quantity would be a mistake in either configuration; what the comparison shows is that the regularised fit parks it at the prior while the unregularised one moves it to -2.264 on no evidence, having started the run with E_f on the -3.0 bound. `Baseline-NN-MSE` puts D on its upper bound of 2.000 in two of its six accepted identifications—the first and the last, so the run both starts and ends on the bound—and E_f on its lower bound of -3.0 in the first. In between it wanders over $D_f \in [1.241, 2.000]$ without settling. A peak factor of 2.000 against a plant that realises 1.05 is not a 90 % error in a tire parameter so much as the box-edge signature of Section III-C: the coefficient is not identified, and nothing in a per-coefficient bounds check distinguishes that from a fit.

Second, the price `Ours` pays is in the linear range. Every slope here is (5) for an identified set and the closed form of Table II for the plant, both at $\bar{F}_z = 1$ and at the same static axle loads $F_{z,f} = 1373$ N, $F_{z,r} = 1274$ N, so the three are comparable. `Ours` gives 13.2 kN/rad front and 13.9 kN/rad rear against the plant's 20.4 and 19.0, i.e. 35 % and 27 % low, while `Baseline-NN-MSE` gives 19.1 and 18.1, 6.5 % and 4.6 % low. The understeer sign is correct in both ($l_r C_r > l_f C_f$), so both sets pass the stability gate of Section VII-A—but not by much in the baseline's case: at the same axle loads it gives $l_f C_f = 14.10$ kN m/rad against $l_r C_r = 14.38$ kN m/rad,

TABLE V
CROSS-RUN COMPARISON OF THE TWO CONFIGURATIONS.

Metric	`Ours`	`Baseline-NN-MSE`
Closed-loop tracking (3 timed laps)		
Timed laps	3	3
Run duration [s]	591.4	564.1
Best lap [s], wall clock	148.60	142.56
Mean lap [s], wall clock	156.85	148.33
RMS e_y, overall [m]	**0.279**	0.303
RMS e_y, pure pursuit [m]	**0.405**	0.406
RMS e_y, MPC [m]	**0.092**	0.177
Max $\lvert e_y \rvert$ [m]	**2.651**	2.660
ITAE e_y [m s^2]	**10362**	17651
IAE e_y [m s]	**50.4**	74.2
IAE/T [m]	**0.085**	0.132
ITAE/$(T^2/2)$ [m]	**0.059**	0.111
RMS heading error [rad]	**0.0195**	0.0197
Max |heading error| [rad]	**0.311**	0.317
Emergency halts	0	0
MPC solve time, RMS [ms]	0.413	**0.390**
MPC solve time, max [ms]	**3.814**	4.061
Axle lateral force against per-wheel telemetry		
Front RMSE [N]	70.68	**45.90**
Rear RMSE [N]	**40.79**	50.06
Front R^2	0.944	**0.978**
Rear R^2	**0.979**	0.970
Front max abs. error [N]	**138**	215
Rear max abs. error [N]	**85**	223
Peak friction against the plant's $\mu = 1.05$		
Front μ RMSE	**0.041**	0.762
Rear μ RMSE	**0.048**	0.772
One-step state prediction		
v_y RMSE [m/s]	0.027	**0.022**
ω RMSE [rad/s]	**0.032**	0.068
v_y persistence RMSE [m/s]	0.009	0.009
ω persistence RMSE [rad/s]	0.005	0.005
Skill vs. persistence, v_y	-2.0	$\mathbf{-1.4}$
Skill vs. persistence, ω	$\mathbf{-5.4}$	-12.6

Bold is the better of the two where the metric has a direction. Lap times are on the ROS wall clock; the two runs did not hold the same real-time factor, so they are reported as recorded and not bolded.
Max $\lvert e_y \rvert$ is the same pure-pursuit out-lap transient in both runs, before either identified model is in the loop, and agrees to 9 mm. ITAE is not scale-free and the runs differ in length, hence IAE/T and ITAE/$(T^2/2)$ beside it.
The persistence rows are the reference baseline—next state equals last measured state—not a result.
Every value is a single run per configuration: no row carries a confidence interval, and the directional rows are one paired observation, not an estimate of a mean difference (Section VIII-C).

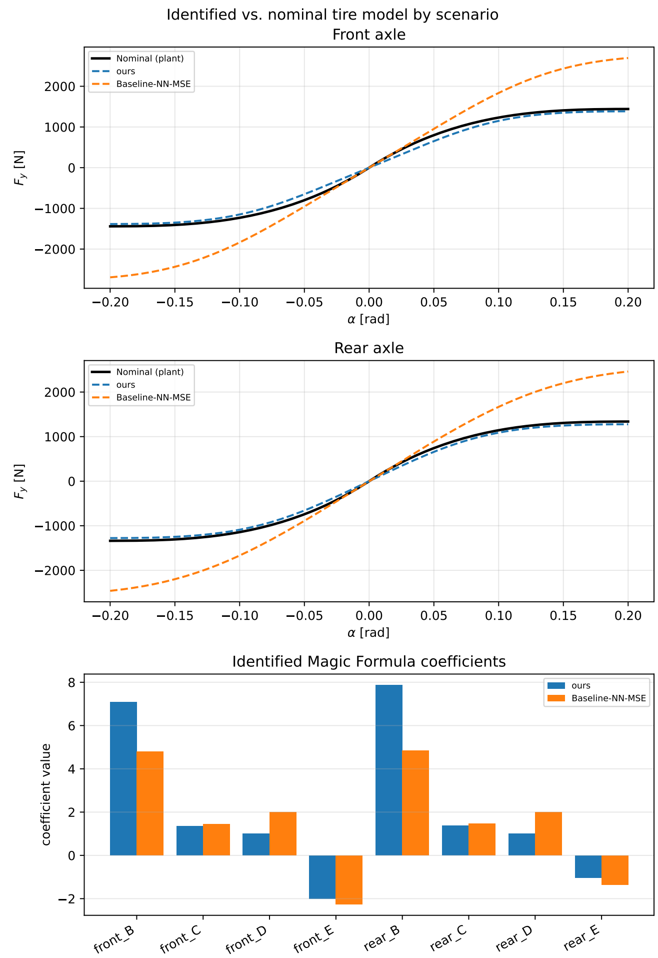


Fig. 5. Identified tire models against the plant. Top and middle: the plant's own curve (8) at the measured static load of each axle (black) against each run's last accepted identification, over $\alpha \in [-0.2, 0.2]$ rad. Bottom: the identified Magic Formula coefficients. `Baseline-NN-MSE`'s last accepted peak factor sits on its box constraint $D = 2.0$ on both axles, so its curve carries roughly twice the plant's force at the edge of the plotted range; `Ours` tracks the plant's peak and undercuts its initial slope.

an understeer margin of 2 %, where `Ours` keeps 12 % (9.74 against 11.06). A set that lands 2 % from an oversteering pair on a coefficient it did not identify is the case the gate exists for, and no per-coefficient bounds check sees it. Of the two sets, only `Baseline-NN-MSE`'s grip ceiling is a claim the plant cannot honour: (6) gives 2.00 g for it against 1.01 g for `Ours` and $\approx 1.0\,g$ measured.

The reason both effects can coexist is the slip-angle coverage of the lap. Over the identification samples of either run, $P_{99}(|\alpha|) = 0.038$ rad (2.2 °) and $\max|\alpha| = 0.045$ rad, against the 12.0 ° at which the plant's curve first reaches its peak (Table II). The measured lap therefore constrains the slope and says almost nothing about the peak; only the synthetic rollout of Section V-B reaches up the curve, and only the warm start of Section V-E3 carries independent information about μ.

E. Peak friction

Table V reports the error of the identified peak factor against the effective friction the plant reports per wheel, and Fig. 6 shows how it evolves over the run. With the surface static at $\mu = 1.05$, `Ours` returns $D_f = 1.009$ and $D_r = 1.002$ in *every* one of its six accepted identifications: the published peak factor is constant for the whole run, so its error is a constant offset and RMSE, MAE and maximum absolute error coincide at 0.041 and 0.048. The warm start's floor does not bind on any cycle here. `Baseline-NN-MSE` spans $[1.241, 2.000]$ front and $[1.358, 2.000]$ rear over its six, starting and ending on the bound, giving RMSE 0.762 and 0.772: 16–19× worse ($0.762/0.041 = 18.6$ front, $0.772/0.048 = 16.1$ rear).

One qualification belongs on the `Ours` column: the corrected pipeline *preserves* a physically valid peak factor rather than sharpening one. The settled value is its configured prior, measured on this vehicle, and the fit does not move it at all—six identifications, six identical coefficient sets to four decimal places. What the comparison shows is that the same data, the same rollout and the same box constraints drive the inherited configuration onto the bound instead, and that the difference is worth 0.99 g of claimed grip.

F. Axle lateral force

The identified curve is scored at the measured (α, F_z) against the plant's per-wheel lateral force; Table V carries the summary, and the bar charts, error histograms and time series behind it are in the supplementary material. The ordering reverses across axles: `Ours` is better at the rear (40.8 N against 50.1 N RMSE, R^2 0.979 against 0.970) and worse at the front (70.7 N against 45.9 N, R^2 0.944 against 0.978). The error histograms are centred in both cases and differ in width and tails: `Baseline-NN-MSE`'s maximum absolute error is 215 N front and 223 N rear against 138 N and 85 N for `Ours`, so its lower front RMSE is bought with a tail 1.6× heavier at the front and 2.6× at the rear. Where the force is largest each model errs in the direction its initial slope predicts—`Ours` short, the inherited configuration long.

This is the linear-range deficit of Section VI-D showing up as force error, and it is the one axis on which the inherited configuration is better. At 2.2 ° of slip both curves are far below their peaks, so a peak factor on its box constraint costs nothing here while a 35 % slope deficit costs 71 N of front-axle RMSE. The metric that separates the two models is therefore not the force at the slip angles the lap visits but the grip ceiling those forces do not probe—which is precisely what a controller planning a speed profile consumes.

G. One-step state prediction

Level 1 of the validation protocol (Section V-G) is reported in Table V against a persistence baseline—next state equals last measured state—written by the same benchmark node. `Ours` predicts yaw rate at 0.032 rad/s RMSE against `Baseline-NN-MSE`'s 0.068 (R^2 0.950 against 0.788), and lateral velocity marginally worse (0.027 m/s against 0.022, R^2 0.988 against 0.992). The error histograms, in the supplementary material, show the yaw-rate difference as tail mass: maximum absolute error 0.452 rad/s for `Ours` against 1.030 rad/s.

Neither model beats persistence at the 0.035 s model step: the skill scores $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{pers}}$ are negative for both configurations and both signals, worst at -12.6 (`Baseline-NN-MSE`, ω) and best at -1.4 (`Baseline-NN-MSE`, v_y). We report this rather than the R^2 alone because R^2 against a slowly varying signal is close to uninformative—a one-step metric on a smooth trajectory rewards not moving, and over one 0.035 s step the plant's own states barely move ($\mathrm{RMSE}_{\mathrm{pers}}$ is 0.009 m/s and 0.005 rad/s). One-step accuracy is consequently not evidence for either tire model here; the axle-force and closed-loop comparisons are. The two state traces against their own ground truth over the whole run are in the supplementary material.

H. Closed-loop tracking

Level 3 of the validation protocol completes on the regenerated trajectory. Both runs finish three timed laps with zero emergency halts and the same finite-state sequence—bootstrap pure pursuit, running pure pursuit, handover, running MPC—with one transition of each kind and no fallback to pure pursuit after handover. The sequence is read off each run's exported state timeline, which records the handover at $t = 81.4$ s for `Ours` and $t = 66.0$ s for `Baseline-NN-MSE` and no state change thereafter; the aggregated `transitions` column of the cross-run summary is not populated and is omitted from Table V. Under MPC, which is where the identified model enters the loop, `Ours` tracks at 0.092 m RMS lateral error against 0.177 m, a 48 % reduction, with maximum 0.490 m against 0.696 m (Fig. 7). Overall ITAE is 10362 against 17651 (supplementary material). ITAE is $\int t|e_y|\,\mathrm{d}t$ and is neither scale-free nor comparable across runs of different length, and these two runs are 591.4 s and 564.1 s long, so it is reported here beside two normalised forms computed from the same samples: mean absolute error IAE/T, 0.085 m against 0.132 m, and $\mathrm{ITAE}/(T^2/2)$, 0.059 m against 0.111 m. All three order the runs the same way and at the same magnitude, so the normalisation does not carry the result. The ITAE split confirms that the whole difference is in the MPC phase: the pure-pursuit phases, which do not use the identified model, give 211 and 122 respectively, ordered the other way and two orders of magnitude smaller. The signed error distribution (supplementary material) shows the mechanism as a narrower distribution about a comparable centre—sample standard deviation 0.279 m against 0.301 m, mean 0.014 m against 0.039 m.

The QP cost is unchanged: RMS solve time 0.413 ms against 0.390 ms, worst case 3.81 ms against 4.06 ms, both two orders of magnitude below the 50 ms control period, and the two orderings disagree, which is what "unchanged" means here. The identified coefficients change the model the QP carries, not its size.

Lap time is not read as a result. Timing runs on the ROS wall clock while the simulator advances its fixed step faster than real time, and the ratio between the two differed between the runs, so the recorded lap times in Table V are not simulated lap times and their 4–6 % spread is not a vehicle-performance difference.

I. Baseline pipeline versus this work

The supplementary material tabulates the pipeline as published [6] against the pipeline as it stands here, on every axis that differs. In summary: the residual model, the data budget, the four-stage loop and the steady-state force recovery are inherited unchanged; what differs is the configuration of the virtual-data step, the regularisation topology, the measurement chain, and the contract at the output. Table III's comparison is a subset of those axes—the rollout correction of Section V-B is active in both runs of Section VI-C, so that section measures the remaining four settings and not the rollout.

J. Peak-friction warm start on a synthetic plant

The friction warm start of Section V-E3 is evaluated on a synthetic single-track plant with brush tires rather than in CARLA, because only a synthetic plant supplies a μ that

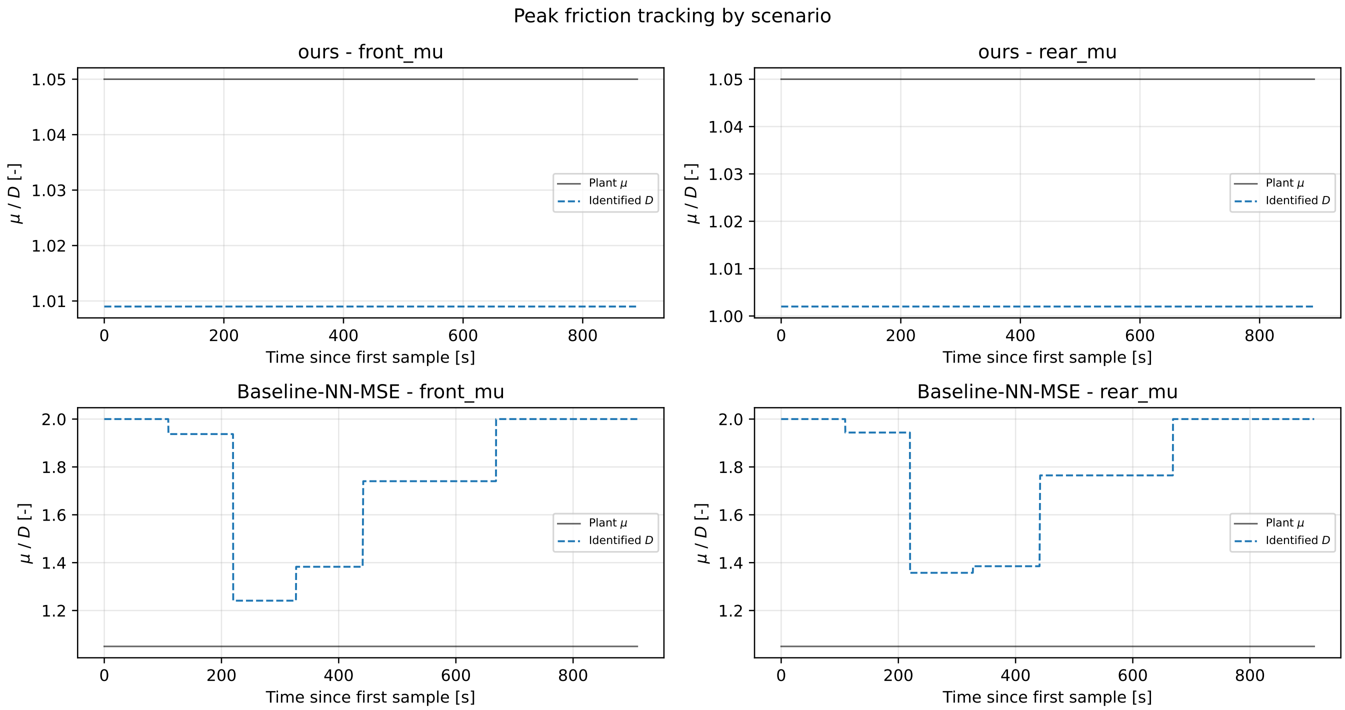


Fig. 6. Peak friction over the run, one row per configuration: the plant's effective μ (black) against the identified peak factor D published at each of the six accepted identifications. The surface is static, so every departure from the black line is estimator error. `Ours` is a flat line about 0.04 below the plant on both axles—the same coefficients every cycle—while `Baseline-NN-MSE` starts and ends on the $D \leq 2.0$ box constraint and does not settle in between.

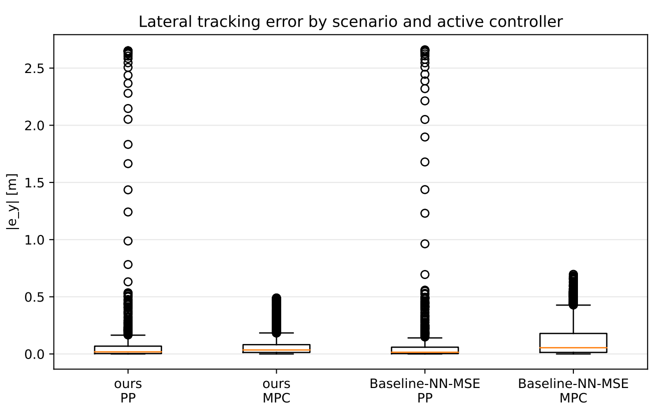


Fig. 7. Absolute lateral tracking error by configuration and active controller. The pure-pursuit boxes cover the acquisition phase before handover and do not use the identified model; the MPC boxes do.

TABLE VI
PEAK FRICTION RECOVERED FROM A SYNTHETIC BRUSH-TIRE PLANT (TRUE $\mu = 1.05$), AGAINST EXCITATION.

δ amplitude	Grip utilisation	Utilisation	Brush fit (13)
0.05 rad	45 %	0.473	**1.048**
0.07 rad	63 %	0.658	**1.048**
0.09 rad	80 %	0.834	**1.048**
0.13 rad	100 %	1.045	**1.048**

The utilisation estimator tracks the grip the manoeuvre *used*; the brush fit of (13) recovers the peak from the curvature of $F_y(\alpha)$ and is therefore flat in excitation.
Below 40 % utilisation the brush fit is gated off and the utilisation bound is reported instead.

is known exactly and an excitation level that can be swept independently of it. The plant is driven by a swept-sine steering input for 40 s at 50 Hz with true $\mu = 1.05$ and $C_\alpha = 12.1F_z$, and the steering amplitude sets how far up the tire curve the manoeuvre reaches. Table VI reports what each estimator returns.

At 45 % utilisation—a level an ordinary lap reaches—the utilisation bound is 55 % low while the brush fit is within 0.2 %. Across true $\mu \in \{0.7, 1.05, 1.4\}$ the fit stays within 5 % and C_α within 10 %; with odometry and IMU noise injected, μ stays within 12 %. Below the 40 % floor the gate suppresses the fit and reports the bound, so the estimator's failure is visible rather than silent. On the CARLA runs above the lap's utilisation leaves the released estimate below the configured prior, so the floor never binds and the warm start does not move D on any of the six cycles (Fig. 6); its contribution there is that it cannot lower D, not that it raises it.

K. Identification cost

One S4D identification—six co-identification iterations on a 30 s lap—takes 20.9 s end to end, inside the 30 s re-identification period the node runs at, and the reduction from the 181.4 s it originally took is exact: no epoch count, iteration count, early-stopping rule or loss weight was changed, and the identified coefficients agree to four decimal places. The per-phase breakdown, the device split, and the caveat that the stage micro-benchmarks and the end-to-end totals are separate measurements rather than a decomposition of each other, are in the supplementary material.

VII. Integration with the Controller

The identified model leaves the pipeline through one interface. A nonlinear MPC path-tracking controller (acados [31], partial-condensing HPIPM [32], horizon $N = 20$ at a 20 Hz control period) carries the same single-track model as the identifier and consumes Φ_p directly: the eight Pacejka coefficients enter the prediction model's axle forces through (4), and nothing else about the controller changes when a new identification arrives. Coefficients are pushed over a ROS 2 service at the end of each identification cycle, so the controller's model tracks the plant at the re-identification rate. We claim no contribution to the controller design; it appears here as the consumer that defines what an admissible identification is.

A. Admissibility gates

A coefficient set can lie strictly inside every box constraint and still be unusable. We observed three distinct ways, each requiring a gate on a *derived* quantity rather than on a coefficient:

1) **Grip ceiling versus trajectory demand.** The controller computes $\max_k(v_{x,k}^2|\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (6) falls below it. On the earlier trajectory the degenerate fit reported a $0.40\,g$ ceiling against a $1.22\,g$ demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a physically plausible set ($D = 1.5$) on the same trajectory, at the same speed and through the same controller. Both figures come from an offline closed-loop harness—20 s on the reference with the handover at $t = 5$ s—and not from a controlled pair of simulator runs; they are quoted as the observation that motivated the gate, not as a result. They are not confounded with the transport-delay correction below, which was already in place when they were measured. The same identified set on the same trajectory gives 0.16 m at 11.1 m/s and 0.43 m at 15 m/s, which is the v^2 scaling the Introduction describes. The gate is one-sided by construction and only catches a ceiling that is too low: `Baseline-NN-MSE`'s $2.00\,g$ (Section VI-D) passes it, since it exceeds every demand the reference makes.
2) **Understeer sign and critical speed.** A set with $l_f C_f > l_r C_r$ whose v_{crit} falls inside the controller's speed range makes the prediction model open-loop unstable, and the resulting $\rho(A_d) > 1$ is raised to ρ^N in the condensed QP. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\mathrm{crit}} = 14.5$ m/s at the measured geometry and mass of Table I, inside the controller's speed range on every trajectory in this paper.
3) **Integration stiffness.** Even a stable set can be stiff relative to the control period: a single explicit RK4 step at $dt = 0.05$ s gave $\rho(A_d) = 547$ at 5 m/s for one identified set. The model sub-steps adaptively, sizing the sub-step from the model's own fastest lateral mode $|\lambda| \approx (C_f + C_r)/(m v_x) + (l_f^2 C_f + l_r^2 C_r)/(I_z v_x)$ so that $|\lambda| h \leq 1.5$, restoring $\rho \approx 1.000$ across the speed range over the shipped $N = 20$ horizon. The cost is carried by the solve: over the reported runs the QP setup and solve took a median of 0.36 ms and a 95th percentile of 0.62 ms against the 50 ms control period.

The gates are the contract between identification and control: an identification that passes is a model the controller can be held responsible for, and one that fails is reported as unidentified rather than silently deployed. Two further controller-side changes were required before any identified model could be evaluated at full-scale speeds and are recorded for completeness rather than as identification results: reference speeds are clamped at ingest, and the solved state is rolled forward by the measured transport delay before each solve, which took cross-track error at 27 m/s from 130.55 m to 0.38 m.

VIII. Discussion and Limitations

A. What generalises

The reachable-friction bound (10) is kinematic: it involves no tire parameter, no simulator-specific quantity and no property of the residual model, so it applies to any implementation of the virtual-steady-state-data strategy of [6], in simulation or on hardware. Its practical content is a design rule—*the rollout must be configured to demand at least the friction one intends to identify*—and, where it cannot, D should be reported as unidentified rather than as fitted. The diagnostic signature, coefficients resting exactly on box constraints, is likewise generic and should be checked and logged by any bounded tire-fitting routine, since it is otherwise indistinguishable in the output from a successful fit.

The insufficiency of per-coefficient box constraints for deployment admissibility is also platform-independent. Both failures we encountered—a set that satisfies every box constraint while being degenerate, and a set that satisfies every box constraint while being pairwise oversteering and unstable within the controller's operating range—are properties of the parameterisation, not of CARLA. Deep Dynamics [21] constrains coefficients inside the network; we argue that a second, derived-quantity check belongs at the interface where the model is handed to a controller.

B. What does not generalise

Three results are properties of this plant and must not be read as statements about tires. First, the numerical tire values: the plant is a PhysX wheeled-vehicle model with a configured friction coefficient, so Table II describes *that* plant, and agreement with it is evidence that the identifier recovers the plant it observes, not evidence about real tire behaviour. Sagmeister *et al.* [28] show that sensitivity to model fidelity is largest near the acceleration limits, which is where this study operates, and R-CARLA [27] exists because CARLA's native dynamics are a known weak point for racing-grade work; hardware replication is therefore necessary before any claim about identification accuracy on a real vehicle. Second, the gap between configured wheel friction (1.5) and realised peak axle friction (1.00–1.05) is an artefact of the server multiplying the configured wheel coefficient by the road surface's own, a measured constant 0.70 on this map; we report it only because an identifier—or a reviewer—tuned to converge on the number in the configuration file will converge on the wrong number, and because the simulator publishes the effective value per wheel, so nothing forces that mistake. Third, the 0.76 achieved/commanded steering ratio arises from CARLA's Ackermann controller and this vehicle's steering configuration—the lesson, identify against the achieved actuator state, is general and standard practice; the magnitude is not.

C. Limitations

1) **One run per configuration, so no metric in Table V carries an uncertainty.** Every value there is a point estimate from a single 591.4 s and a single 564.1 s run. The 0.092 m against 0.177 m RMS tracking difference, in particular, has no confidence interval attached and no evidence that it exceeds run-to-run variation, which is not measured here. Nothing about the experiment prevents repetition—the sweep is one command and the scenario file already carries the per-arm parameter sets—so this is a gap in the evidence rather than in the method.
2) **Single surface, vehicle and trajectory.** All results come from one vehicle configuration on one map with the friction held static at 1.05. The runtime friction override supports a surface-change adaptation experiment—arguably the strongest argument for on-track identification—but it is not performed here.
3) **The friction estimator's excitation sweep is synthetic** (Table VI). The synthetic plant was chosen because μ is known there and excitation can be swept independently of it. On the CARLA runs the warm start is exercised only at the one excitation level the lap happens to supply, and only through the floor it puts under D on the first identification, so its excitation dependence is not verified against the simulator's per-wheel forces.
4) **The two configurations differ in four settings at once** (Table III), so Section VI attributes nothing to the residual architecture, the loss, the solver or the warm start individually; in particular we do not claim to reproduce the S4 improvement of [22]. The excitation-aware rollout—the paper's central contribution—is active in *both* arms, so the closed-loop comparison of Section VI-H is a comparison of the four secondary settings and is not evidence for the identifiability correction. The evidence for that correction is the rollout-speed sweep of Section VI-B. What is claimed about the S4D residual on its own is its cost and the exactness of the reduction (Section VI-K), a statement about the implementation rather than about the tire it identifies.
5) **Longitudinal dynamics are out of scope.** As in the baseline only lateral dynamics are identified; combined slip and load transfer are neglected, and the simulator's longitudinal wheel force channel is drivetrain torque rather than a contact-patch force, so it is unusable as ground truth for a longitudinal extension without further work.

IX. Conclusion and Future Work

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [6] at full vehicle scale, using a CARLA simulation architecture that exposes the plant's own per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, the pipeline fails in a specific, diagnosable and correctable way. The failure is not a solver, bounds or tuning problem: it is a loss of excitation created by the pipeline's own virtual-data step, whose reachable friction $\mu_{\text{reach}} = v^2\delta_{\text{sweep}}/(gL)$ depends on the rollout configuration and not on the tire. A rollout speed appropriate at 1:10 caps μ_{reach} at 0.43 at full scale, below the plant's peak, so the peak factor is not identifiable from the synthetic data and the bounded optimiser resolves the ambiguity on a box edge.

Correcting this—together with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved rather than the commanded road-wheel angle, measured mass and geometry, and a friction warm start identified from the curvature of the axle force curve and applied only as a floor—produces identifications with no railed coefficient and a peak factor of 1.009/1.002 against a plant realising 1.05, with the correct understeer sign. Run against the inherited configuration over three timed laps on a static surface, it *preserves* that peak factor to 0.041 and 0.048 RMSE where the inherited configuration starts and ends on its 2.0 box constraint, and halves the MPC's lateral tracking error, from 0.177 m to 0.092 m RMS, at unchanged solver cost. Preserves, not identifies: this trajectory never approaches the tire's peak, the accepted fits return the configured prior to four decimal places, and the reported peak-friction accuracy is therefore the prior's (Section VI-A4). That the peak factor is recoverable *when the excitation reaches it* is shown on a known tire in Section VI-B, not on these laps. Each configuration is a single run, so the closed-loop margins are observations rather than estimates with a confidence interval. The inherited configuration is better on front-axle force and on lateral-velocity prediction, because the lap reaches 2.2 ° of slip and a wrong peak factor is not observable there—which is the same identifiability statement seen from the metric side. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities at that interface instead; the earlier reference trajectory, which demanded 1.22 g from a vehicle sustaining 1.0 g, was regenerated because that gate refused it rather than loosened for it.

A. Future work

a) Immediate:

1) Separate the four settings of Table III. The comparison reported here moves the residual architecture, the objective, the solver and the warm start together, so it establishes that the shipped configuration is better on the metrics that matter to a controller without saying which

change earns it. The runtime objection to the temporal arm is removed (Section VI-K).

2) Excite the peak. Every result here is obtained at $P_{99}(|\alpha|) = 2.2\,^\circ$ against a plant whose curve peaks at $12.0\,^\circ$, which is why a peak factor on its box constraint costs almost nothing in axle force. A trajectory or an injected manoeuvre that reaches further up the curve would make the two models separable on force as well as on friction.
3) Run the noise sweep of [6] on this platform using the bridge's odometry noise model, with a nonlinear least-squares identification of the same data as the comparison arm.
4) Cross-check the friction warm start against the simulator's per-wheel forces across a pace sweep, including the σ_μ/μ gate boundary and the IMU-versus-kinematics cross-check that guards the mass and inertia assumptions. It is so far validated against a synthetic brush-tire plant, where μ is known exactly (Table VI).
5) Report lap time on the simulator's own clock. Lap timing currently runs on the ROS wall clock against a faster-than-real-time simulator, which makes lap time unusable as a comparison metric (Section VI-H).
6) Measure I_z rather than assuming it (Section A).

b) Beyond:

1) **Excitation-aware data collection.** Equation (10) inverts into a requirement on the manoeuvre. A controller that deliberately inserts bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would move the pipeline from passive on-track identification toward active, safety-bounded on-track experiment design—which is optimal input design [52] under a safety constraint, and is where the literature for this step already exists.
2) **Uncertainty-aware output.** The gates of Section VII-A are binary. Reporting a posterior over the coefficients—the Bayesian variational path already implemented alongside the deterministic solvers—would let the controller weight the model by its confidence rather than accept or reject it.
3) **Hardware replication.** The decisive test of every claim in Section VIII is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, which is what makes the simulation study a necessary precursor rather than a substitute.

Appendix A
Assumptions and Reproducibility Notes

The following are assumptions or design choices rather than measured facts. They are listed explicitly so that a reader can judge how far each result depends on them, and so that each can be discharged by a specific measurement.

A1. Yaw inertia I_z

$I_z = 51.1\,\mathrm{kg\,m^2}$ is carried in the model file and is *not* measured. It enters (2) directly and therefore scales the yaw-rate residual the network is trained on, and it enters the stiffness-adaptive sub-stepping of Section VII-A through $l_f^2 C_f + l_r^2 C_r$ over I_z. *How to discharge:* apply a known yaw moment (a step steer at constant speed with the resulting axle forces read from the force feedback topic) and fit I_z from the measured $\dot{\omega}$; or read the value the physics engine is using directly from the actor's inertia tensor.

A2. Centre-of-gravity split

$l_f = 0.738142\,\mathrm{m}$, $l_r = 0.795362\,\mathrm{m}$ are taken from the controller configuration and confirmed by the vehicle owner. The measured static load split (51.2/48.8 front/rear) implies a marginally different split than the geometry does (51.9/48.1) — a 1.4 % disagreement. We have not resolved it, and it is below the level at which any conclusion in this paper would change.

A3. CG height and load transfer

$h_{\mathrm{cg}} = 0.3\,\mathrm{m}$ is carried in the model file and unused: the single-track model of Section III-A neglects load transfer, following the baseline. Normal loads are static. On a full-scale vehicle at $1\,g$ this is a stronger approximation than at 1:10, and it is a candidate explanation for part of the residual the network absorbs.

A4. Steering-angle sign and units

`feedback/steering_angle` is published in degrees and is treated as the front road-wheel angle, positive left, matching the ROS body frame. It was verified to agree with the front-left wheel joint state to within 0.5 %. Left and right road wheels are not distinguished; the single-track δ is taken as this single reported value.

A5. Steady-state force recovery

Equation (7) is not a force measurement — it is fully determined by $v_x\omega$ — so it misattributes transients. It is applied only to the *synthetic*, quasi-steady rollout, where the steady-state assumption is constructed rather than hoped for. Its adequacy was nonetheless checked on measured data against the simulator's own axle force (correlation 0.993, slope 0.94); we treat that as validating the relation, not as validating a steady-state assumption on recorded laps.

A6. The residual network is not the plant

The corrected model $\hat{\mathbf{x}}_{k+1} + e_k$ is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section V-C limits but does not eliminate this: the sweep still reaches $1.5\times$ the observed 99th percentile of $|\delta|$. The choice of 1.5 is empirical, selected because it was sufficient to reach the tire's peak at the corrected rollout speed while removing the observed drift; it is not derived.

A7. Simulator determinism and timing

CARLA runs in synchronous mode with a 0.033 s physics step, but sim time advances faster than wall time (observed $\approx 1.9\times$ on our hardware). All identification data is stamped from the simulation clock.

A8. Software configuration of record

Identification and control workspace: ROS 2 Humble, Python 3.10 (required — ROS 2 Humble ships CPython 3.10 ABI extensions only), PyTorch with CUDA on an RTX 2080-class GPU. The bridge is built and sourced as a separate overlay on top of that workspace, so that its message packages (`sim_manager_msgs`) resolve for the ground-truth forces; CARLA server 0.9.16 / Unreal Engine 4.26; **acados 0.5.5** with partial-condensing HPIPM. The identifier's acquisition and state-machine timer runs at 50 Hz (20 Hz is the controller's period) with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at 5×10^{-4}. The one-step model is discretised at $T_s = 0.035$ s (`nn_params.yaml: sample_dt`). This is an empirical value—it matches neither the server's 0.033 s step nor the 50 Hz acquisition timer—and is carried because the identification is insensitive to it over the range tested (Section IV-E).

Random seeds. `nn_train` seeds PyTorch and NumPy at 42 for the single-member architectures used here (42+i per ensemble member). `pacejka_solver.seed` is `null` in both shipped parameter sets, so the multi-start jitter and—for `Ours`—the differential-evolution population draw from an unseeded generator: the reported identifications are reproducible in distribution but not bit for bit. The sweep of Table IV is seeded explicitly and is bit-reproducible.

APPENDIX B
SIMULATOR–ROS 2 INTERFACE

Command path

The bridge runs in `ackermann_drive` mode, subscribing to a single `ackermann_msgs/AckermannDriveStamped` on `/drive` and forwarding speed and steering angle to CARLA's native Ackermann controller, whose server-side cascaded PID loops are configured once at startup. No client-side steering PID is present—the commanded road-wheel angle passes through in radians—so the achieved/commanded ratio of Table I is a property of the vehicle and its steering curve rather than of a tuning choice in the bridge.

Feedback path

Table VII lists the interface consumed by the identifier and its validation. One property of it is specific to this architecture.

a) Per-wheel ground truth: `feedback/tire_forces` carries CARLA's own `WheelTelemetryData`: per-wheel lateral slip angle, normal load, lateral force, effective friction coefficient, wheel speed, longitudinal force and wheel torque, in wheel order [FL, FR, RL, RR]. No analytic tire model exists inside the bridge, so nothing on this topic can disagree with the physics the vehicle is driven by. This topic is what allows an on-track identifier to be scored against the forces the plant generated. Four properties of it were established by measurement and bound what may be read off it. (i) Only `lateral_force` is force ground truth; it is the channel every validation in this paper uses. (ii) `longitudinal_force` is a friction-capacity report rather than a measurement—it tracks μF_z and correlates with ma_x at 0.04—and `wheel_torque` is an exact identity of it, so neither carries independent information. (iii) Below 0.5 m/s the publisher zeroes slip angle and both forces while the normal load stays live, so a load-only gate admits standstill frames as perfect $(0, 0)$ samples; the identifier gates on the all-zero pattern as well. (iv) The slip angle needs no sign change across the boundary, which was verified rather than assumed.

TABLE VII
ROS 2 INTERFACE USED BY THE IDENTIFICATION AND VALIDATION PIPELINE. NAMESPACE IS `/SIM` UNLESS MARKED ABSOLUTE.

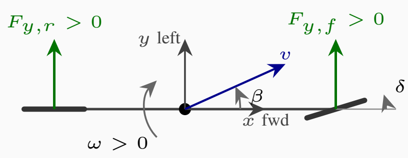
Topic	Type	Role
`/odom`	`nav_msgs/Odometry`	State v_x, v_y, ω; twist is body-frame
`feedback/steering_angle`	`std_msgs/Float32` (deg)	Achieved road-wheel angle δ
`feedback/imu`	`sensor_msgs/Imu`	a_x, a_y for the friction warm-start
`feedback/tire_forces`	`sim_manager_msgs/TireForces`	Per-wheel α, F_z, F_y, μ, ω_w, κ: **ground truth**
`feedback/vehicle_physics`	`std_msgs/String` (JSON)	Live wheel parameters of (8)
`feedback/speed`	`std_msgs/Float32`	Forward speed echo
`feedback/motors`	`std_msgs/String` (JSON)	Per-wheel speed/torque/brake
`/drive`	`ackermann_msgs/AckermannDriveStamped`	Control command in
`control/tire_friction`	`std_msgs/Float32`	Runtime grip override
`/clock` *(abs.)*	`rosgraph_msgs/Clock`	Simulation time base

Figure 8 states the conventions the rest of the paper assumes. The simulator and the identification stack do not share a handedness: CARLA inherits Unreal's left-handed world (X east, Y *south*, Z up, yaw positive clockwise), while every ROS 2 topic follows REP-103 [53]—right-handed ENU for the world, forward-left-up for the body—and the identification equations of Section V are written in the body frame with α and F_y positive to the left, the SAE-style axis convention [54], [4]. The bridge is where the two meet, and it is a per-channel conversion rather than a single transform: position and yaw flip sign, twist is additionally rotated into the body frame by q^{-1}, the IMU's a_y flips, the steering *command* flips on the way out, and the tire telemetry's slip angle does *not*, because the frame flip and CARLA's own sign convention for `lat_slip` cancel. That last one is the trap: a sign error there is invisible in the states and inverts the identified curve, so it was checked against a slip angle rebuilt from the chassis state in a steady 4 m/s turn at 0.20 rad—equal magnitude to within 2 % front and 15 % rear.

Extensions made for identification

Six changes support this study and are listed for reproducibility: the existing but unsubscribed `feedback/steering_angle` is now consumed by the identifier (with configurable units, scale and staleness timeout, falling back to the `/drive` setpoint); runtime overrides `control/tire_friction` and `control/drag_coefficient` write straight to the live `VehiclePhysicsControl`, so surface friction can change within a run without respawning; the torque curve, zero-throttle damping and wheel

(a) Vehicle body frame (FLU)



$$\alpha_f = \delta - \arctan \frac{v_y + l_f \omega}{v_x}, \quad \alpha_r = - \arctan \frac{v_y - l_r \omega}{v_x}$$

Positive α produces positive (leftward) F_y. Wheel order in every per-wheel array: [FL, FR, RL, RR].

(b) Conversions applied by CARLA_ASU_BRIDGE

Channel	**Bridge conversion**
`/odom` pose	$y \mapsto -y$; ψ, $\theta \mapsto -\psi, -\theta$
`/odom` twist	world $\rightarrow$ body by q^{-1}; v_y, ω_y, ω_z negated
`feedback/imu`	$a_y \mapsto -a_y$
`tire_forces` α	deg $\rightarrow$ rad only: the frame flip and CARLA's own sign convention cancel
`tire_forces` F_y	$F_y \mapsto -F_y$, into $+y$ left
`/drive` δ	negated on the way out to CARLA

Fig. 8. Frame and sign conventions across the simulator–ROS 2 boundary. (a) All identification quantities are defined in the body frame: x forward, y left, ω and δ positive counter-clockwise, and slip angle and lateral force positive to the left, so that a positive α produces a positive F_y [54]. (b) The conversion applied per channel. The slip angle of the tire telemetry is the one exception that needs no sign change, because the frame flip and CARLA's own convention cancel.

`tire_friction` were tuned against a low-speed hold test (a wheel friction above ≈2.0 on a steered wheel scrubs hard enough at creep speed to stall the drivetrain intermittently), and these values define the plant being identified; telemetry runs on a dedicated 10 Hz thread off the control loop, since each tick costs several blocking CARLA RPCs; `feedback/tire_forces` was extended with the effective friction coefficient, the wheel speed and a kinematically recomputed slip ratio $\kappa = (\omega_w r_w - v)/v$, the last because CARLA's own `long_slip` field is inconsistent with the wheel speed and radius the same message reports (+0.177, i.e. hard braking, on a wheel turning 8.99 rad/s at a steady 2.33 m/s cruise, whose kinematic ratio is -0.034); and the live wheel parameters of (8) are published on `feedback/vehicle_physics`, so the reference curve of Table II cannot go stale when the vehicle configuration changes.

APPENDIX C
REFERENCE IMPLEMENTATION

The identification package, the ROS 2 bridge and the controller stack used in this paper are available as `On-Track-SysID`, extended from [6] (upstream: https://github.com/ForzaETH/On-Track-SysID), CARLA_ASU_BRIDGE (https://github.com/ebrahimabdelghfar/Carla_ASU_Bridge) [39], and the accompanying MPC and benchmarking packages, which are released together with the scenario definitions, the exported result CSVs and the figure and sweep generators at https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo/tree/feat-follow-premade-racing-track. The benchmarking packages compute one-step and multi-step prediction metrics online (RMSE, MAE, NRMSE, MaxAE, bias, standard deviation and R^2 by Welford's algorithm), compare estimated against ground-truth per-wheel lateral forces, and record controller-switching behaviour, and were built for this study.

ACKNOWLEDGMENT

The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.* [6], released under an open-source licence; this paper reports a port, a correction, and an extension of that work rather than a reimplementation from scratch.

FUNDING

This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ETHICS

All results in this paper are obtained in simulation. No experiment involved human subjects, personal data, or the operation of a physical vehicle, so no human-subjects approval and no vehicle-safety protocol were required.

DATA AND CODE AVAILABILITY

The identification package, the ROS 2 bridge, the MPC and the benchmarking packages, the scenario definitions, the exported result CSVs behind every figure and table of Section VI, and the sweep generator behind Table IV are in the repositories listed in Appendix C; the top-level project repository is https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo/tree/feat-follow-premade-racing-track.

ON-TRACK-SYSID is an extension of the ForzaETH release of [6] and is distributed under the MIT licence it declares in its package manifest; CARLA_ASU_BRIDGE is GPL-3.0. The two are used as separate ROS 2 overlays communicating only over ROS 2 topics and services, and neither links against the other, so no combined-work obligation arises between them; the identification and control packages in this work are released under the MIT terms the ON-TRACK-SYSID upstream requires.

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